

Modelling Financial Markets

Chapter 2: Market Efficiency — From EMH to Adaptive Markets



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Acronyms used in this material (1/2)

- **AI**: Artificial Intelligence
- **AIC**: Akaike Information Criterion
- **AMH**: Adaptive Markets Hypothesis
- **AR**: AutoRegressive (model); in event studies: Abnormal Return
- **BDS**: Brock–Dechert–Scheinkman (test)
- **BET**: Bucharest Exchange Trading (index)
- **BIC**: Bayesian Information Criterion
- **BIST**: Borsa İstanbul (BIST 100 index) — *the Istanbul Stock Exchange (BIST 100 index)*
- **BNR**: Banca Națională a României — *the National Bank of Romania*
- **BUX**: Budapesti Értéktőzsde index — *the Budapest Stock Exchange index*
- **BVB**: Bursa de Valori București — *the Bucharest Stock Exchange*
- **CAPM**: Capital Asset Pricing Model
- **CD**: Chow–Denning (multiple variance-ratio test)
- **CI**: Confidence Interval
- **CME**: Chicago Mercantile Exchange
- **CRSP**: Center for Research in Security Prices
- **DFA**: Detrended Fluctuation Analysis
- **EMH**: Efficient Market Hypothesis
- **ETF**: Exchange-Traded Fund
- **fGn**: fractional Gaussian noise
- **FRED**: Federal Reserve Economic Data
- **FTSE**: Financial Times Stock Exchange (FTSE Russell, index provider)
- **FWER**: Family-Wise Error Rate

Acronyms used in this material (2/2)

- **FX**: Foreign eXchange
- **GARCH**: Generalised AutoRegressive Conditional Heteroskedasticity
- **GMM**: Generalized Method of Moments
- **HAC**: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- **HK**: Hong Kong
- **IPC**: Índice de Precios y Cotizaciones — *the Mexican Stock Exchange index*
- **MDE**: Minimum Detectable Effect
- **MDH**: Martingale Difference Hypothesis
- **MIT**: Massachusetts Institute of Technology
- **ML**: Machine Learning
- **MSPE**: Mean Squared Prediction Error
- **NYSE**: New York Stock Exchange
- **OLS**: Ordinary Least Squares
- **PX**: Prague Stock Exchange index (index Burzy cenných papírů Praha)
- **RW1**: Random Walk 1 (i.i.d. increments)
- **RW2**: Random Walk 2 (independent increments)
- **RW3**: Random Walk 3 (uncorrelated increments)
- **SD**: Standard Deviation
- **SE**: Standard Error
- **SPA**: Superior Predictive Ability (test)
- **TOM**: Turn Of the Month
- **TSMOM**: Time-Series MOMentum
- **US**: United States
- **VR**: Variance Ratio
- **WIG20**: Warszawski Indeks Giełdowy 20 — *the Warsaw Stock Exchange index of the 20 largest companies*

Chapter Route and Outcomes

The question of this chapter

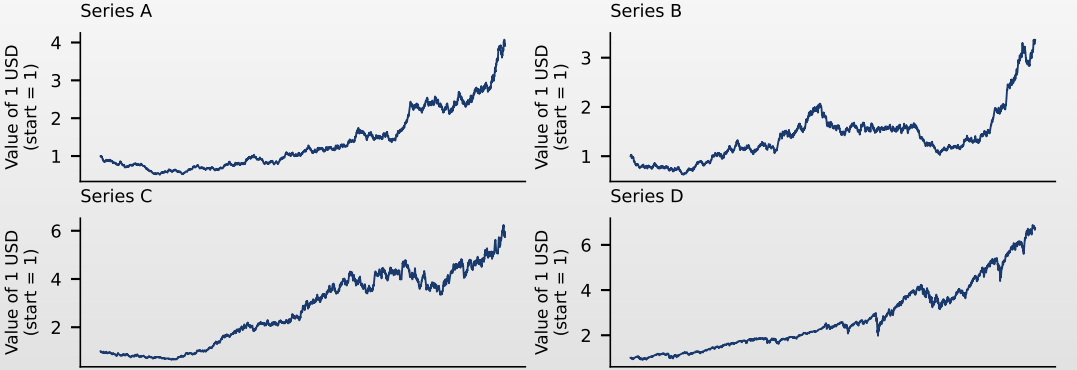
- If past prices revealed a strategy with positive expected risk-adjusted profit after costs, traders would exploit it until it disappeared
- A market in which this has already happened is called **efficient**; the claim restricts the statistical behaviour of returns, so it can be tested
- Part I: definitions, weak-form tests, predictive regressions, long memory
- Part II: adaptive markets, event studies, anomalies, behavioural finance, machine learning, a case study and one pass of the AI-assisted research loop; technical detail in the ▸ Appendix

After this chapter you can

- state which information set a test uses and which null it actually tests (RW1, RW3, martingale difference)
- run autocorrelation, variance-ratio, long-memory and calendar tests with inference robust to volatility clustering and to multiple testing
- separate a statistical rejection from an exploitable profit and from an irrational market

A Game: Which Series Is the Real Market? (1/2)

One of these is the S&P 500 (2010-2026); three are simulated random walks with the same drift and volatility



A Game: Which Series Is the Real Market? (2/2)

- Vertical axis: P_t/P_0, the value reached by 1 USD invested on the first day (for the S&P 500: the price index, without dividends)
- One panel is the S&P 500 (2010–2026); three are random walks with the same mean and volatility. Vote: A, B, C or D?

MFM_ch2_random_walk_game

A Game: Which Series Is the Real Market? (2/2)

- Vertical axis: P_t/P_0, the value reached by 1 USD invested on the first day (for the S&P 500: the price index, without dividends)
- One panel is the S&P 500 (2010–2026); three are random walks with the same mean and volatility. Vote: A, B, C or D?
- Answer: **D**. Roberts (1959): random walks produce "patterns" by chance; we need statistics, not eyes
- Bachelier (1900) and Samuelson (1965): properly anticipated prices fluctuate randomly, so unpredictability is the mark of a well-functioning market

MFM_ch2_random_walk_game

Data Used in This Chapter

- 16 daily series up to 18 September 2026; sample sizes: Appendix
  - 13 equity indices from 2000: 4 developed (S&P 500, Euro Stoxx 50, Nikkei 225, Hang Seng), 9 emerging or frontier (BET, WIG20, BUX, PX, BIST 100, Bovespa, IPC, Nifty 50, Shanghai)
  - Bitcoin (from 2014; from 2011 in the yearly crypto analysis), Ethereum (from 2016) and EUR/RON, BNR reference rate (from July 2005)
- Daily log returns $r_t = \ln P_t - \ln P_{t-1}$, each series on its own calendar
  - P_t: the closing price (index level, exchange rate) on day t; ln: the natural logarithm
- Equity indices: weekdays only; a close equal to the previous close is removed, since it is usually a holiday filled with the previous price
  - the rule also removes the rare genuine zero-return day (S&P 500: 3 of about 6 700 days); the test statistics move by less than 0.05
- Crypto-assets trade 7 days a week: 365 observations per year
- Analyses that combine several series (e.g. event studies) align prices on common trading days before computing returns

Efficiency and the Information Set

- Fama (1970): a market is **efficient** if prices "fully reflect" all available information $\mathcal{I}_t$; portrait and surveys: Appendix
  - formally: no strategy using $\mathcal{I}_t$ earns **expected** returns above the fair (risk-adjusted) return

Form	Information set $\mathcal{I}_t$	Typical test (this chapter)
Weak	Past prices and returns	Autocorrelation, runs, variance ratios, long memory
Semi-strong	All public information (news, reports)	Event studies
Strong	Public and private (insider) information	Returns of insiders and fund managers

- Nested: strong $\Rightarrow$ semi-strong $\Rightarrow$ weak; risk-adjusted returns with factor models follow in Chapter 3

Efficiency as a Fair Game: the Martingale (1/3)

- **Fair game** for excess returns: $\mathbb{E}[r_{t+1} - \mu_t \mid \mathcal{I}_t] = 0$
  - $\mathcal{I}_t$: the information available at the end of day t; $\mathbb{E}[\cdot \mid \mathcal{I}_t]$: the expectation given that information
  - μ_t: the required (equilibrium) return, the compensation for risk; in words: the return beyond μ_t cannot be forecast from $\mathcal{I}_t$
- **Martingale**: a process whose expected next value, given the past, equals its current value
  - With a constant required simple return m, the total-return price discounted at m, $P_t/(1+m)^t$, is a martingale Campbell, Lo & MacKinlay (1997); discounting at the risk-free rate gives a martingale only under the risk-neutral measure
  - P_t itself is one only if $m = 0$: a zero expected **simple** return, not a zero expected log return

Efficiency as a Fair Game: the Martingale (2/3)

- "$\mathbb{E}[P_{t+1} \mid \mathcal{I}_t] = P_t$" and "$\mathbb{E}[\ln(P_{t+1}/P_t) \mid \mathcal{I}_t] = \mu$" are different restrictions
  - They differ by discounting and by the Jensen term: $\mathbb{E}[\ln(1+R)] \approx \mathbb{E}[R] - \mathrm{Var}(R)/2$ for small returns ($\mathbb{E}[R]^2$ neglected)
  - $R = P_{t+1}/P_t - 1$: the simple return; $\ln(1+R)$: the log return; the mean log return is lower than the mean simple return by about half the variance
- Consequence: $\mathrm{corr}(\varepsilon_{t+1}, g(\mathcal{I}_t)) = 0$ for any function g of the past, $\varepsilon_{t+1} = r_{t+1} - \mu_t$
  - ε_{t+1}: the unexpected part of the return; in words, no past information is correlated with tomorrow's surprise
  - In particular, all autocorrelations $\rho_k = \mathrm{corr}(r_t, r_{t-k})$, $k \geq 1$, are zero when $\mu_t = \mu$; k: the lag in days

Efficiency as a Fair Game: the Martingale (3/3)

- With a constant required simple return m, the fair game is $\mathbb{E}[P_{t+1} + D_{t+1} \mid \mathcal{I}_t] = (1+m)\,P_t$; D_{t+1}: the cash flow (dividend) paid in $t+1$
  - Fundamental value $P_t^* = \mathbb{E}\big[\sum_{j\geq 1} D_{t+j}/(1+m)^j \mid \mathcal{I}_t\big]$: the present value of the expected cash flows at the required return; pricing error $B_t = P_t - P_t^*$
  - P_t^* satisfies the same recursion; subtracting: $P_t = P_t^* + B_t$ with $\mathbb{E}[B_{t+1} \mid \mathcal{I}_t] = (1+m)\,B_t$
- A fair game ties the price to the fundamental value only up to a **rational bubble**
  - A gap $B_t \neq 0$ is admissible only if it is expected to grow at the required rate m (Chapter 17)
  - A mean-reverting gap (fad) makes ε_{t+1} predictable and violates the fair game, although short-horizon tests have little power against it Summers (1986)
- The restriction is on the conditional mean only: the variance is not restricted, and volatility may be predictable (Chapter 1)

Random Walk Hypotheses RW1, RW2, RW3

- Log price $p_t = \ln P_t$ with $p_t = \mu + p_{t-1} + \varepsilon_t$; the three versions differ in what they assume about ε_t Campbell, Lo & MacKinlay (1997)
  - μ: the drift, the expected daily log return; ε_t: the shock of day t, with mean 0; so $r_t = p_t - p_{t-1} = \mu + \varepsilon_t$
- **RW1**: ε_t i.i.d., independent and identically distributed (the strongest; rules out volatility clustering)
  - Rejected by the stylised facts of Chapter 1
- **RW2**: ε_t independent but not identically distributed (variance may change)
- **RW3**: ε_t uncorrelated: $\mathrm{Cov}(\varepsilon_t, \varepsilon_{t-k}) = 0$, while $\mathrm{Cov}(\varepsilon_t^2, \varepsilon_{t-k}^2) \neq 0$ is allowed
  - The relevant hypothesis for real returns: tests must be robust to heteroskedasticity

The Joint-Hypothesis Problem

- "Fair return" requires a model of expected returns (e.g. CAPM, Chapter 3)
- Every test of efficiency is a **joint** test of efficiency and of the asset-pricing model Fama (1991)
  - A rejection may mean an inefficient market **or** a wrong model of risk
  - Predictable **expected** returns are allowed if they compensate time-varying risk
- Some investors beat the market by luck: the question is whether success persists
- At daily horizons expected returns are tiny, so the choice of model matters little: weak-form tests are the cleanest
- Efficiency can never be "proved", only not rejected by a given test on given data

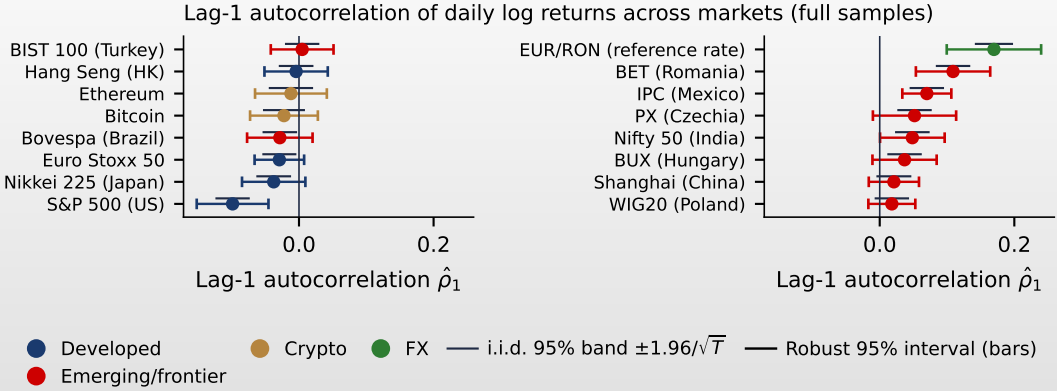
The Grossman–Stiglitz Paradox

- Gathering information is costly (analysts, data, computing)
- If prices reflected **all** information, nobody would be paid for collecting it Grossman & Stiglitz (1980)
  - Then nobody would collect it, and prices could not reflect it
- Resolution: an **equilibrium degree of inefficiency**, just large enough to reward informed traders for their costs
- Markets with cheap information and low costs (large, liquid) should be closer to efficiency: a prediction for our comparison of developed, emerging and crypto markets
- Nor does efficiency rule out bubbles: it says they are hard to time (Chapter 17)

Autocorrelation and Its Robust Standard Error

- Sample autocorrelation: $\hat{\rho}_k = \sum_{t=k+1}^{T} e_t e_{t-k} / \sum_{t=1}^{T} e_t^2$, with $e_t = r_t - \bar{r}$
  - T: number of daily returns; $\bar{r}$: their mean; k: the lag; $\hat{\rho}_k > 0$: persistence, $\hat{\rho}_k < 0$: reversal
- Under RW1 (i.i.d.): $\mathrm{SE}(\hat{\rho}_k) \approx 1/\sqrt{T}$; 95% band $\pm 1.96/\sqrt{T}$
- Unpredictable mean with volatility clustering (finite fourth moments): $\mathrm{SE}(\hat{\rho}_k) \approx \sqrt{\hat{\tau}_k}$
  - $\hat{\tau}_k = \sum_t e_t^2 e_{t-k}^2 \,/\, \left(\sum_t e_t^2\right)^2$ (robust variance)
  - When large moves cluster, $\hat{\tau}_k > 1/T$: the i.i.d. band is too narrow and classical tests reject too often; every test below uses a variance that allows for this
- S&P 500: i.i.d. SE $= 0.012$, robust SE $= 0.027$, more than twice as large

Lag-1 Autocorrelation Across Markets (1/2)



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Lag-1 Autocorrelation Across Markets (2/2)

- Dots: $\hat{\rho}_1$ with robust 95% intervals; thin segments: the naive i.i.d. band; colours: the market group (FX, foreign exchange: EUR/RON)
- Positive and significant: BET 0.109 (SE 0.028), IPC 0.070, EUR/RON 0.170
- Negative and significant: S&P 500 -0.099
- Bitcoin -0.022, Ethereum -0.012: not significant
- Candidate explanations: thin trading (BET), a managed exchange rate (EUR/RON), short-term overreaction and liquidity provision (S&P 500)
- Economic size: $R^2 = \hat{\rho}_1^2$, i.e. 0.97% (S&P 500) and 1.19% (BET) of the daily variance

MFM_ch2_autocorrelation

Portmanteau Tests: Ljung–Box and Its Robust Version

- Ljung & Box (1978): $Q(m) = T(T+2)\sum_{k=1}^{m} \hat{\rho}_k^2/(T-k) \sim \chi^2_m$ under i.i.d. returns
  - tests H_0: $\rho_1 = \cdots = \rho_m = 0$ jointly; m: number of lags; χ^2_m: chi-squared distribution with m degrees of freedom; a large Q rejects
- Robust version Lobato, Nankervis & Savin (2001): $\tilde{Q}(m) = \sum_{k=1}^{m} \hat{\rho}_k^2/\hat{\tau}_k \to \chi^2_m$ under RW3 when the $\hat{\rho}_k$ are asymptotically uncorrelated (condition on the slide The Heteroskedasticity-Robust Statistic z^*)
- Three markets at $m = 10$ (5% critical value 18.31)
  - S&P 500: $Q = 96.3$ but $\tilde{Q} = 19.9$ ($p = 0.031$)
  - BET: $Q = 108.0$, $\tilde{Q} = 24.4$ ($p = 0.007$)
  - Bitcoin: $Q = 20.2$ ($p = 0.028$) but $\tilde{Q} = 11.9$ ($p = 0.294$)
- The robust test keeps the rejections for S&P 500 and BET, but not for Bitcoin

The Runs Test

- Nonparametric: uses only the **signs** of returns (zero returns dropped), so it is immune to heavy tails Wald & Wolfowitz (1940)
- A **run** is a maximal block of equal signs: ++|-|+++ has 3 runs; Seminar A5 counts them on 20 signs
- Moments of the number of runs R under independence, with n_1 positive and n_2 negative signs, $n = n_1 + n_2$
  - $\mathbb{E}[R] = \dfrac{2n_1n_2}{n} + 1, \qquad \mathrm{Var}[R] = \dfrac{2n_1n_2(2n_1n_2 - n)}{n^2(n-1)}, \qquad z = \dfrac{R - \mathbb{E}[R]}{\sqrt{\mathrm{Var}[R]}} \approx N(0,1)$
  - R: the observed number of runs; $\mathbb{E}[R]$, $\mathrm{Var}[R]$: its mean and variance if the signs were independent; $|z| > 1.96$ rejects at 5%
- Too **few** runs ($z < 0$): signs persist; too **many** ($z > 0$): sign reversals; the test assumes a constant probability of a positive day
- Three markets: S&P 500 $z = 4.34$ (more runs, as $\hat{\rho}_1 < 0$); BET $z = -6.05$ (fewer runs, as $\hat{\rho}_1 > 0$); Bitcoin $z = 2.68$ although $\hat{\rho}_1$ is not significant
  - signs and magnitudes carry different information; all markets: Appendix

The Variance Ratio: Idea

- Uncorrelated increments with constant variance: variance grows linearly with the horizon, $\mathrm{Var}(r_t + \cdots + r_{t-q+1}) = q\,\mathrm{Var}(r_t)$
  - q: the horizon in days (e.g. $q = 5$ is one trading week)
  - With changing variances (RW2, RW3) the variance of the sum is the sum of the variances; the sample $\widehat{\mathrm{VR}}(q)$ still tends to 1 under the conditions of Lo & MacKinlay (1988)
- **Variance ratio**: $\mathrm{VR}(q) = \dfrac{\mathrm{Var}(r_t + \cdots + r_{t-q+1})}{q\,\mathrm{Var}(r_t)}$, equal to 1 for any covariance-stationary uncorrelated series
- For a covariance-stationary series, a weighted sum of autocorrelations: $\mathrm{VR}(q) = 1 + 2\sum_{k=1}^{q-1}\left(1 - \frac{k}{q}\right)\rho_k$
- $\mathrm{VR}(q) > 1$: momentum (positive autocorrelation); $\mathrm{VR}(q) < 1$: mean reversion
- Worked check: S&P 500 $1 + \hat{\rho}_1 = 0.901$ vs estimated $\mathrm{VR}(2) = 0.901$

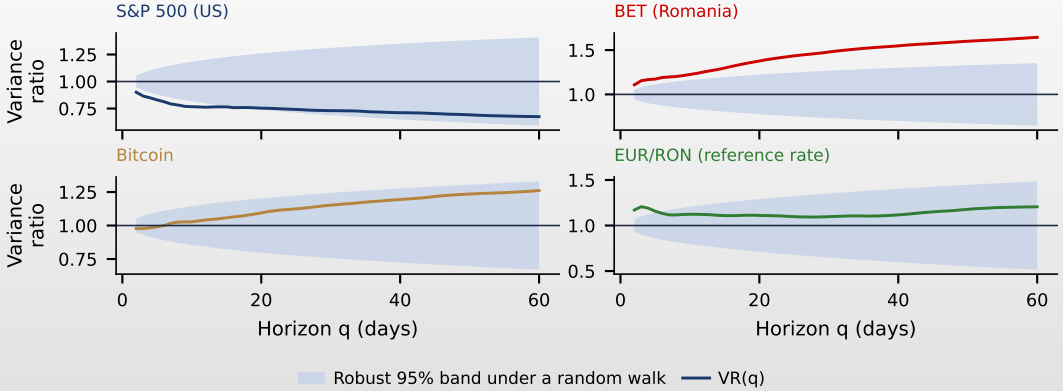
The Lo–MacKinlay Test

- Lo & MacKinlay (1988): overlapping q-day sums, with bias corrections
  - $\hat{\sigma}_a^2 = \frac{1}{T-1}\sum_t (r_t - \hat{\mu})^2, \quad \hat{\sigma}_c^2(q) = \frac{1}{m}\sum_{t=q}^{T} \left(\sum_{j=0}^{q-1} r_{t-j} - q\hat{\mu}\right)^2$
  - $m = q(T-q+1)(1-q/T); \quad \widehat{\mathrm{VR}}(q) = \hat{\sigma}_c^2(q)/\hat{\sigma}_a^2$
  - $\hat{\mu}$: mean daily return; $\hat{\sigma}_a^2$: variance of one-day returns; $\hat{\sigma}_c^2(q)$: variance of the overlapping q-day sums divided by q; m: a divisor that removes the small-sample bias
- Test statistic under RW1 (homoskedastic returns)
  - $z(q) = \dfrac{\widehat{\mathrm{VR}}(q) - 1}{\sqrt{2(2q-1)(q-1)/(3qT)}} \approx N(0,1)$
  - the denominator is the standard error of $\widehat{\mathrm{VR}}(q)$ under RW1; $|z| > 1.96$ rejects $\mathrm{VR}(q) = 1$ at 5%
- Homoskedastic: constant variance; this z over-rejects when volatility clusters (next slide)

The Heteroskedasticity-Robust Statistic z^*

- Under RW3 plus the conditions of Lo & MacKinlay (1988) (finite moments, weak dependence (mixing), $\mathbb{E}[\varepsilon_t^2 \varepsilon_{t-j}\varepsilon_{t-k}] = 0$ for $j \neq k$), the $\hat{\rho}_k$ are asymptotically uncorrelated with variances $\hat{\delta}_k / T$
  - $\hat{\delta}_k = T\hat{\tau}_k$, with $\hat{\tau}_k$ of the autocorrelation slide: the robust variance of $\sqrt{T}\hat{\rho}_k$, equal to 1 without volatility clustering
- Asymptotic variance of $\widehat{\mathrm{VR}}(q)$ and the test statistic
  - $\hat{\theta}(q) = \sum_{k=1}^{q-1} \left(\dfrac{2(q-k)}{q} \right)^2 \hat{\delta}_k, \qquad z^*(q) = \dfrac{\sqrt{T}\left(\widehat{\mathrm{VR}}(q) - 1\right)}{\sqrt{\hat{\theta}(q)}} \approx N(0,1)$
  - $2(q-k)/q$: the weight of $\hat{\rho}_k$ in $\widehat{\mathrm{VR}}(q)$; $|z^*| > 1.96$ rejects $\mathrm{VR}(q) = 1$ at 5%
- RW3 alone is not enough: the conditions hold for GARCH with symmetric innovations, but can fail with a leverage effect
  - Without the last condition, $\hat{\theta}(q)$ needs the full covariance matrix of the $\hat{\rho}_k$
- S&P 500, $q = 2$: $\widehat{\mathrm{VR}} = 0.901$, $z = -8.12$ but $z^* = -3.64$: still significant, but half as large in absolute value (Seminar B1 works it step by step)

Variance-Ratio Profiles (1/2)



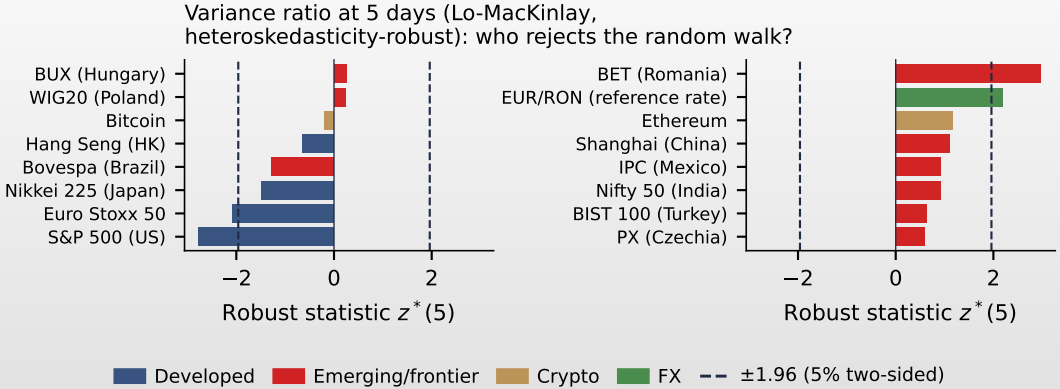
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Variance-Ratio Profiles (2/2)

- $\widehat{\mathrm{VR}}(q)$ for $q = 2, \ldots, 60$; shaded: robust 95% band around 1, from $\hat{\theta}(q)$
- S&P 500: $0.901, 0.830, 0.753, 0.674$ at $q = 2, 5, 20, 60$: reversal, outside the band at short horizons only
- BET: $1.108, 1.173, 1.377, 1.644$: persistence accumulating with q
- Bitcoin: inside the band up to about one month; EUR/RON: short-run persistence of a managed float
- The band widens with q: a curve can point in one direction without being significant; direction is not evidence

MFM_ch2_variance_ratio

Several Horizons and Markets: the Chow–Denning Test (1/2)



MFM_ch2_variance_ratio

Several Horizons and Markets: the Chow–Denning Test (2/2)

- Bars: robust $z^*(5)$ for all 16 series; dashed lines ± 1.96
- Single horizon, reject at 5%: BET (2.98), EUR/RON (2.19), S&P 500 (-2.77), Euro Stoxx 50 (-2.08)
- Four horizons at 5% each: false-rejection probability up to $1 - 0.95^4 = 18.5\%$
  - Chow & Denning (1993): CD $= \max_i |z^*(q_i)|$, the largest of the four $|z^*|$, over $q_i = 2, 5, 10, 20$; 5% critical value 2.49, above 1.96 to pay for the four looks
- The joint test rejects RW3 for 5 of the 16 series: BET, S&P 500, IPC, EUR/RON, Ethereum; Bitcoin $p = 0.835$
- No ranking "developed = efficient": the direction differs (reversal vs persistence)

MFM_ch2_variance_ratio

Beyond Linear Predictability: the Martingale Difference Hypothesis

- **MDH**: $\mathbb{E}[r_t \mid \mathcal{I}_{t-1}] = \mu$: no function of the past predicts the mean, linear or not
- RW3 (all $\rho_k = 0$) is weaker. Counterexample: $r_t = \varepsilon_t + \theta(\varepsilon_{t-1}^2 - \sigma^2)$, ε_t i.i.d. symmetric
  - $\sigma^2 = \mathrm{Var}(\varepsilon_t)$, so the extra term has mean 0; $\theta \neq 0$: how strongly yesterday's squared shock moves today's mean
  - all $\rho_k = 0$, yet $\mathbb{E}[r_t \mid \varepsilon_{t-1}] = \theta(\varepsilon_{t-1}^2 - \sigma^2)$ is predictable: autocorrelation tests cannot see it
  - Tests of the MDH itself: the generalised spectral test Escanciano & Velasco (2006); an automatic portmanteau Escanciano & Lobato (2009); details and results: Appendix
- **BDS** Brock, Dechert, Scheinkman & LeBaron (1996) tests i.i.d., not the MDH: GARCH alone makes it reject
  - A BDS rejection is evidence against RW1, not against weak-form efficiency

Predictive Regressions and the Stambaugh Bias (1/2)

- Question: does a valuation ratio known today forecast next period's return?
- Model: $r_{t+1} = \alpha + \beta x_t + u_{t+1}, \quad x_{t+1} = \theta + \rho x_t + v_{t+1}$, with $\mathrm{Cov}(u, v) = \sigma_{uv}$
  - r: excess return; x: log dividend–price ratio $dp_t = \ln D_t - \ln P_t$, persistent ($\rho \approx 1$)
  - D_t: dividends paid over the last year; β: the predictive slope, 0 if returns are unpredictable; α, θ: intercepts
  - ρ: the persistence of x; u, v: the shocks of the two equations; σ_{uv}: their covariance; $\sigma_v^2 = \mathrm{Var}(v)$; T: number of periods

Predictive Regressions and the Stambaugh Bias (2/2)

- x_t is predetermined, not strictly exogenous: OLS is consistent but biased Stambaugh (1999)
  - Project u on v: $u = (\sigma_{uv}/\sigma_v^2)\, v + e$, so $\hat{\beta} - \beta = (\sigma_{uv}/\sigma_v^2)(\hat{\rho} - \rho) + \text{noise}$; e: the part of u unrelated to v
  - $\hat{\rho}$ is biased downward in small samples Kendall (1954), and this error passes into $\hat{\beta}$
  - With $\mathbb{E}[\hat{\rho} - \rho] \approx -(1 + 3\rho)/T$: $\quad \mathbb{E}[\hat{\beta} - \beta] \approx -\dfrac{\sigma_{uv}}{\sigma_v^2}\,\dfrac{1 + 3\rho}{T}$
- Price up $\Rightarrow$ *dp* down: $\sigma_{uv} < 0$, so the bias is **upward**: predictability looks stronger than it is
- Near a unit root the t-statistic is not $N(0,1)$ either: valid inference by a Bonferroni Q-test over a confidence interval for ρ Campbell & Yogo (2006)

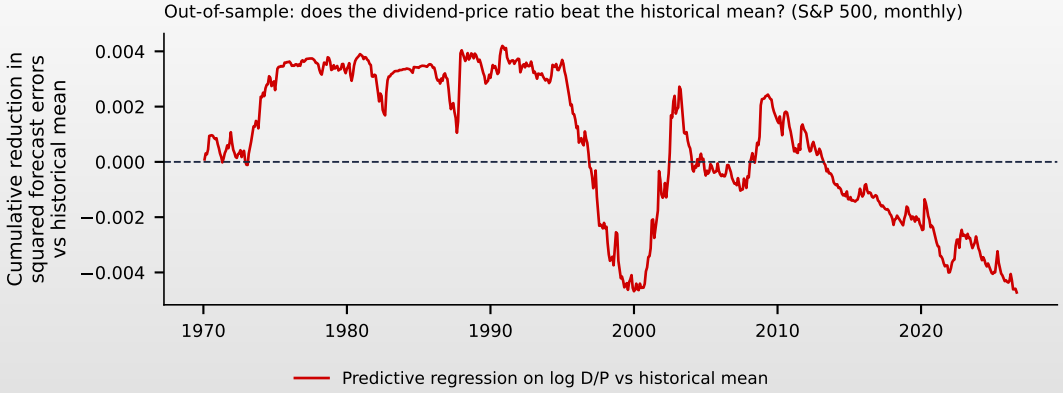
Worked Example: the Stambaugh Bias for the S&P 500

- Monthly log excess return regressed on the lagged dp_t, 1934–2026, $T = 1\,111$ months; data construction: Appendix
- OLS: $\hat{\beta} = 0.267$ (% per month per unit of dp), Newey–West $t = 0.91$: not significant even before any correction
- Inputs: $\hat{\rho} = 0.9988$, $\text{corr}(\hat{u}, \hat{v}) = -0.97$, $\hat{\phi} = \hat{\sigma}_{uv}/\hat{\sigma}_v^2 = -0.925$ (as fractions, not in %)
- Substitution: bias $= -100\,\hat{\phi}\,(1 + 3\hat{\rho})/T = 0.333$ (in %, like $\hat{\beta}$)
  - corrected slope $= 0.267 - 0.333 = -0.066$: the whole estimate is bias
  - the correction is itself estimated: formal inference by Campbell & Yogo (2006) or a bootstrap
- Simulation with the same ρ and $\text{corr}(u, v)$, $\beta = 0$, $T = 360$ months: the nominal 5% t-test rejects $\beta = 0$ in 26.5% of samples

Out-of-Sample Predictability: R^2_{OS} and the Clark–West Test

- Recursive forecasts: at each month t, re-estimate on data up to $t-1$ only (expanding window) and forecast $\hat{r}_t$
  - benchmark: the historical mean $\bar{r}_t$ of the same data; $R^2_{OS} = 1 - \sum_t (r_t - \hat{r}_t)^2 / \sum_t (r_t - \bar{r}_t)^2$; $R^2_{OS} > 0$: the predictor beats the mean
  - Most predictors have $R^2_{OS} < 0$ Welch & Goyal (2008); the 2022 update confirms it Goyal, Welch & Zafirov (2024)
  - Pseudo out-of-sample, as in Welch & Goyal (2008): monthly dividends are interpolated from quarterly totals, so dp_t is not strictly real-time
- Nested models: under H_0 the larger model estimates a zero β, which inflates its MSPE (mean squared prediction error); the Diebold–Mariano test Diebold & Mariano (1995) then rejects less often than its nominal level
  - Clark & West (2007): $f_t = (r_t - \bar{r}_t)^2 - \left[(r_t - \hat{r}_t)^2 - (\bar{r}_t - \hat{r}_t)^2\right]$; one-sided t on the mean of f_t
  - f_t: the squared-error gain of the predictor, plus the term $(\bar{r}_t - \hat{r}_t)^2$ that removes the noise of estimating β; a positive mean favours the predictor
- Campbell & Thompson (2008): impose the sign of β and a non-negative forecast

Out-of-Sample Evidence: the Dividend–Price Ratio (1/2)



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Out-of-Sample Evidence: the Dividend–Price Ratio (2/2)

- Cumulative squared-error gain over the historical mean, forecasts from 1970; rising line: *dp* helps
- 680 forecasts: $R^2_{OS} = -0.51\%$; the Campbell–Thompson restrictions bind in 5.7% of months and leave R^2_{OS} practically unchanged, at -0.51%
- Clark–West $t = 0.42$ (one-sided $p = 0.34$): no significant out-of-sample predictability
- The gain peaks in 1990 and is lost in the 1990s boom, when *dp* fell to record lows; after 2010 it erodes again
- Economic relevance: with monthly Sharpe ratio 0.160, even $R^2 = 0.5\%$ would raise by 20% the expected return of a mean–variance investor who times the market with the forecast; formula:

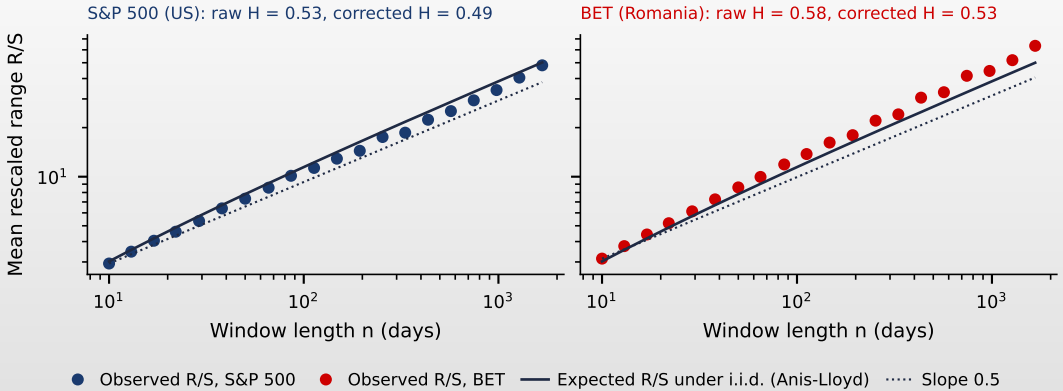
▸ Appendix

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The Rescaled Range (R/S) Statistic

- Using the rescaled range, Hurst (1951) found that wet years on the Nile followed wet years more often than chance allows; history: Appendix
- Construction for a window of n returns with mean $\bar{r}_n$ and standard deviation s_n
  - Cumulative deviations $Y_k = \sum_{i=1}^{k}(r_i - \bar{r}_n)$, $k = 1, \ldots, n$
  - Range $R_n = \max_k Y_k - \min_k Y_k$; rescaled range R_n/s_n
- Scaling law: $\mathbb{E}[R_n/s_n] \approx c\, n^H$, where H is the **Hurst exponent**
  - c: a constant; taking logs, $\ln(R/S) \approx \ln c + H \ln n$, so H is a slope
  - Estimate: slope of $\ln(R/S)$ on $\ln n$, averaging R/S over non-overlapping windows
- $H = 0.5$: no long memory (short-run dependence allowed); $H > 0.5$: persistence; $H < 0.5$: anti-persistence
  - **Long memory** ($H > 0.5$): autocorrelations decay so slowly that their sum diverges, $\rho_k \sim k^{2H-2}$

R/S Analysis: S&P 500 and BET (1/2)



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R/S Analysis: S&P 500 and BET (2/2)

- Dots: average R/S by window length n; solid line: the Anis & Lloyd (1976) expectation under i.i.d. returns from the Normal distribution; exact formula: Appendix
- Small-sample bias: even for i.i.d. returns the raw slope exceeds 0.5; the Anis–Lloyd corrected exponent $H_{AL} = 0.5+$ slope of $\ln(R/S) - \ln \mathbb{E}[R_n/s_n]$
- S&P 500: raw 0.530, corrected 0.486: the "persistence" was the artefact
- BET: raw 0.578, corrected 0.533: persistence remains
- Open: is BET's $H > 0.5$ **long** memory or just $\hat{\rho}_1 = 0.109$? Classical R/S cannot tell

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Hurst Exponent and Variance Ratio: One Horizon Is Not Enough

- For exactly self-similar increments (fractional Gaussian noise): $\mathrm{Var}(r_t + \cdots + r_{t-q+1}) = q^{2H}\mathrm{Var}(r_t)$
- Hence $\mathrm{VR}(q) = q^{2H-1}$ and the **implied** $H(q) = 0.5 + \ln \mathrm{VR}(q)/(2 \ln q)$: BET 0.553, S&P 500 0.453 at $q = 20$
- Short-memory counterexample: an AR(1) with $\rho_1 = -0.1$, $\rho_k = (-0.1)^k$
  - $\mathrm{VR}(20) = 0.826$, so $H(20) = 0.468$: "anti-persistent" without any long memory
  - yet $H(250) = 0.482$: for short memory $H(q) \to 0.5$ as $q \to \infty$ (Seminar A2–A3)
  - an AR(1) with BET's $\hat{\rho}_1 = 0.109$ gives $H(20) = 0.535$: most of the gap between BET's $H(20)$ and 0.5 is short memory
- Separating the two needs a statistic built for it: Lo's modified R/S (next slide)

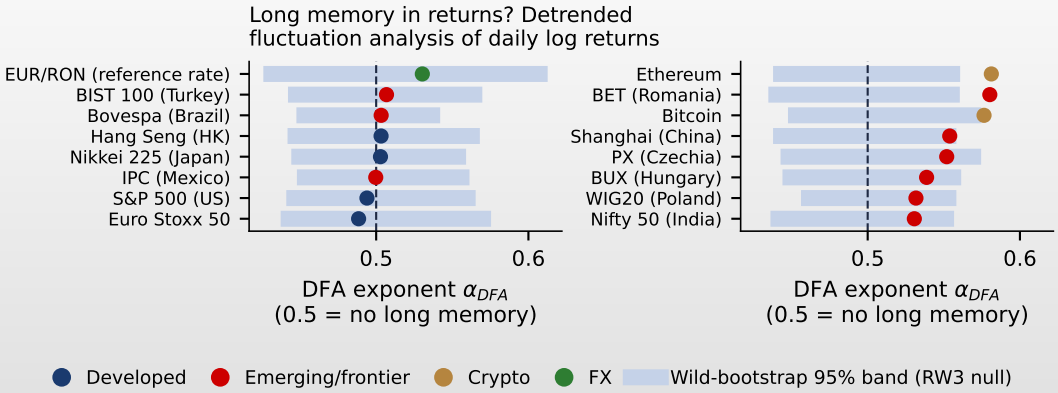
Lo's Modified R/S Test

- Lo (1991): short-run autocorrelation inflates R/S and mimics long memory
- Modification: s_n is replaced by a long-run (HAC) standard deviation $\hat{\sigma}_\ell$ with ℓ lags
  - $\hat{\sigma}_\ell^2 = \hat{\gamma}_0 + 2\sum_{j=1}^{\ell}\left(1 - \frac{j}{\ell+1}\right)\hat{\gamma}_j$, $\hat{\gamma}_j$: autocovariance at lag j ($\hat{\gamma}_0$: the variance), with declining weights
  - $V = \dfrac{\max_k Y_k - \min_k Y_k}{\hat{\sigma}_\ell \sqrt{T}}$; Y_k: cumulative deviations over the full sample of T returns; bandwidth ℓ chosen as in Andrews (1991)
- Under short memory, V lies in $[0.809, 1.862]$ with probability 95%
- Full samples: S&P 500 $V = 1.559$, Bitcoin 1.547, PX 1.750, EUR/RON 1.429: short memory is not rejected
  - BET $V = 1.961$ ($\ell = 7$) and Ethereum $V = 1.980$: just above 1.862, weak evidence of long memory

Detrended Fluctuation Analysis (DFA)

- Peng et al. (1994): linear detrending of the profile removes a locally constant mean of returns; trends in returns need higher-order DFA
- Algorithm in three steps (Seminar B5 shows one window step by step)
  - Profile y_k: the cumulative deviations Y_k of R/S, over the full sample; split into boxes of length n
  - In each box fit a linear trend and keep the residuals
  - $F(n)$ = root mean square of all residuals; repeat for many n
- $F(n) \propto n^{\alpha_{DFA}}$: the DFA exponent α_{DFA} plays the role of H ($\alpha_{DFA} = 0.5$: no long memory)
- Null band by **wild bootstrap**: $e_t \eta_t$, $\eta_t = \pm 1$ with probability $1/2$ (199 replications); η_t: a random sign
  - Keeps the volatility path $|e_t|$ (allowed under RW3) and destroys serial correlation; shuffling would destroy both
  - Implicit null: signs symmetric and independent given $|e_t|$, stronger than RW3: a benchmark, not an exact RW3 test and not a confidence interval for α_{DFA}

DFA Exponent Across Markets (1/2)



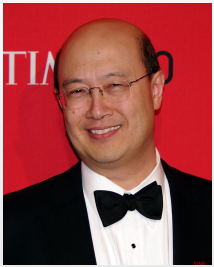
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DFA Exponent Across Markets (2/2)

- Dots: α_{DFA}; bars: sign-flip 95% null band (volatility kept, no autocorrelation)
- Inside the band: S&P 500 0.494, Euro Stoxx 50 0.488, Nikkei 0.503, PX 0.552, Shanghai 0.554
- Above (3 of 16): Ethereum 0.581, BET 0.580 (band to 0.561), Bitcoin 0.576, borderline (band to 0.575)
- A shuffled band flags 5: PX and Shanghai were "memory" only through volatility clustering
- Cross-checks: Lo's V rejects short memory marginally for BET and Ethereum, not for Bitcoin; the local Whittle estimator of the long-memory parameter $d = H - 0.5$ for BET: $\hat{d} = 0.118$, 95% CI $[0.062, 0.174]$; details: Appendix
- Lesson: memory in returns is weak and fragile; memory in volatility (Chapter 1) is strong

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The Adaptive Market Hypothesis



Andrew W. Lo, MIT

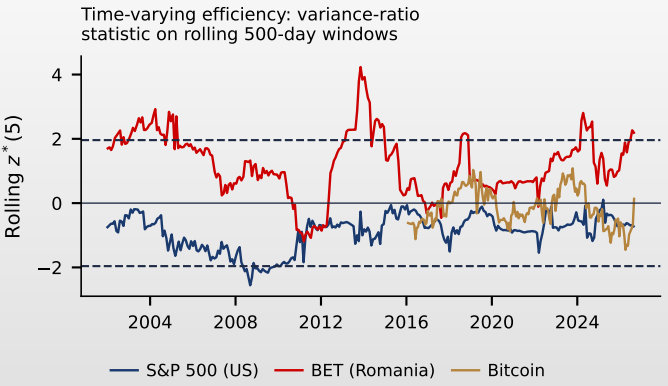
Photo: David Shankbone (2012); CC BY 3.0; Wikimedia Commons

- Lo (2004); book-length version Lo (2017)
- Markets are an ecology of competing species (traders, funds, algorithms) that adapt by trial and error, not by perfect optimisation
  - efficiency is a matter of degree and depends on the number of competitors, the profit opportunities and the environment
- Prediction: efficiency **varies over time and with market conditions**; anomalies appear, weaken and may reappear
- Caution: time-varying statistics alone do not identify the AMH, since an EMH with time-varying risk premia also allows them; comparison table: ▸ Appendix
- Evidence of time variation: survey Lim & Brooks (2011); long-run indices Urquhart & McGroarty (2016)

Measuring Time-Varying Efficiency

- Compute a statistic on a **rolling window**: 500 trading days (about two years), moved forward by 21 days (one month)
  - Statistics: $z^*(5)$ of the variance-ratio test and the DFA exponent
- Null band for DFA: wild bootstrap **inside each window** (199 sign flips), so the band follows the window's volatility
  - An i.i.d. Student-t_4 band $[0.378, 0.614]$ ignores volatility clustering (allowed under RW3), which raises α_{DFA}
- Two cautions for reading the results
  - Overlapping windows are strongly correlated: runs of rejections are not independent evidence
  - Even under the null about 5% of windows reject; a share near 5% is a description, not a calibrated test of stability

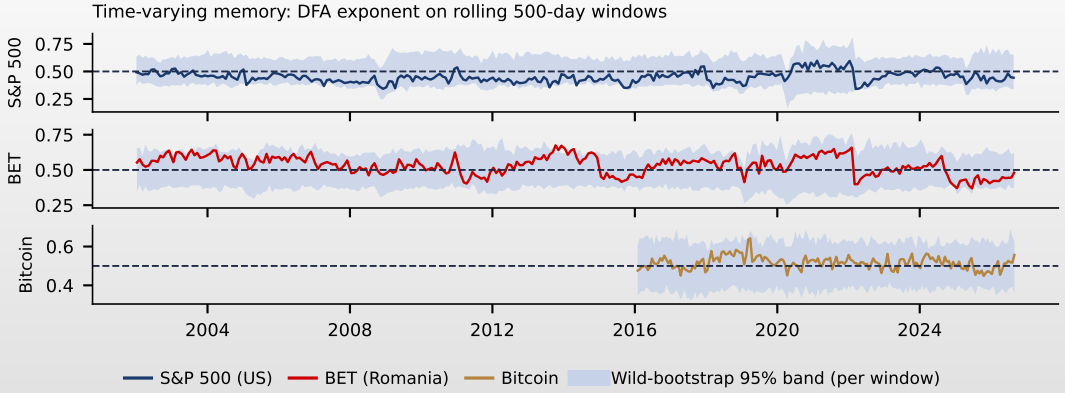
Rolling Variance Ratio: S&P 500, BET, Bitcoin



- $z^*(5)$ on rolling 500-day windows; dashed lines ± 1.96
- Share of windows with $|z^*(5)| > 1.96$: S&P 500 6.4%, BET 22.4%, Bitcoin 0.0%
- S&P 500: the full-sample rejection sits in a few turbulent windows
- BET: rejections cluster in episodes, then fade; FTSE upgrade to Secondary Emerging effective 21 Sep 2020, a candidate change in the investor base; Seminar C2 and ▸ Appendix
- Bitcoin, 2014–2026: no rejection in any window

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Rolling DFA Exponent: S&P 500, BET, Bitcoin (1/2)



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Rolling DFA Exponent: S&P 500, BET, Bitcoin (2/2)

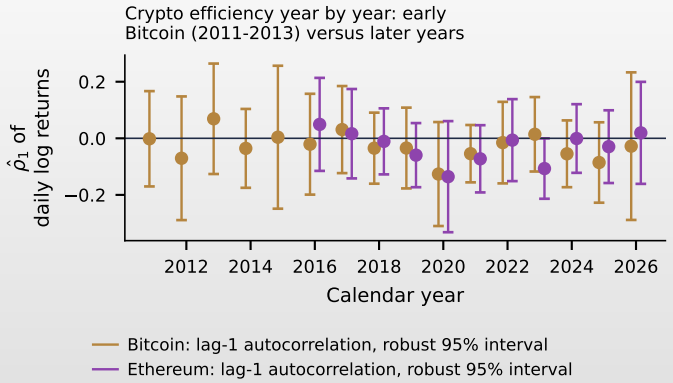
- α_{DFA} on rolling 500-day windows; shaded: wild-bootstrap 95% band computed in each window
- Outside their own band: S&P 500 2.0%, BET 2.7%, Bitcoin 0.0% of windows
- With the fixed i.i.d. Student-t_4 band: 6.4%, 12.9%, 1.1%: the apparent BET "episodes" were volatility clustering
- Low power: the BET band averages $[0.36, 0.64]$; 500 observations cannot pin down a scaling exponent

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The AMH, Tested

- Kim, Shamsuddin & Lim (2011): century-long US data; predictability varies with market conditions (crises, bubbles), not along a trend
  - This turns the AMH narrative into a regression with testable coefficients
- Our version: regress the rolling $z^*(5)$ on the window's annualised volatility (%) and a crisis dummy (window overlaps Sep 2008–Jun 2009 or Feb–Jun 2020)
  - Newey–West errors with 24 lags: consecutive windows share 95% of their days
- S&P 500: crisis -0.67 ($t = -2.37$), volatility 0.006 ($t = 0.27$), $R^2 = 0.20$
- BET: crisis -0.82 ($t = -2.77$), volatility -0.008 ($t = -0.36$), $R^2 = 0.15$
- Crisis windows are associated with lower $z^*(5)$ (towards reversal) in both markets, beyond volatility: a conditional association, not a causal effect
  - z^* mixes $\widehat{\mathrm{VR}} - 1$ and its standard error; state dependence is consistent with the AMH but does not identify it uniquely

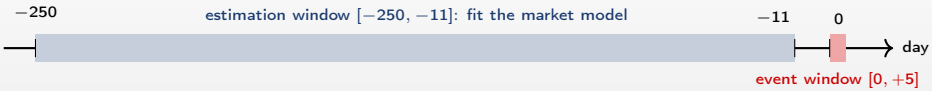
Crypto Efficiency Year by Year



- Lag-1 autocorrelation for each calendar year, robust 95% intervals; Bitcoin from 2011
- Only 1 of 27 yearly $\hat{\rho}_1$ has $|t| > 1.96$ (Ethereum 2023, -0.107), against $0.05 \times 27 = 1.35$ expected by chance
- Unlike Urquhart (2016), who rejects efficiency in the early years: other tests (including BDS), another price source and classical errors; literature:
  ▸ Appendix
- Power: one year of 365 returns gives a robust SE of about 0.06–0.13, so small effects are invisible

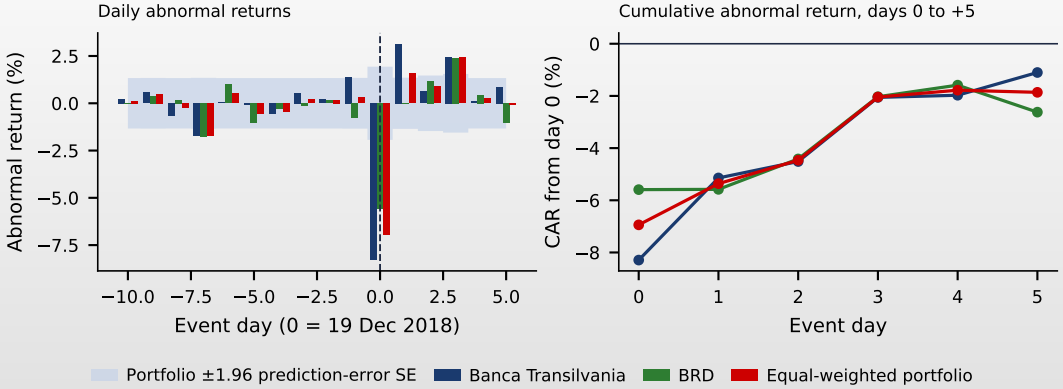
MFM_ch2_crypto_efficiency

The Event-Study Method



- Fama, Fisher, Jensen & Roll (1969): how fast do prices absorb public news (stock splits)?
- Market model: $r_{i,t} = a_i + b_i r_{m,t} + \varepsilon_{i,t}$, $r_{m,t}$ the market return; abnormal return $AR_{i,t} = r_{i,t} - \hat{a}_i - \hat{b}_i r_{m,t}$; cumulative abnormal return $CAR_i[\tau_1, \tau_2] = \sum_{t=\tau_1}^{\tau_2} AR_{i,t}$
- First-pass test: $t = CAR_i / (\hat{\sigma}_\varepsilon \sqrt{L})$, L the event-window length MacKinlay (1997)
  - a_i, b_i: intercept and market beta of firm i, fitted in the estimation window; $\hat{\sigma}_\varepsilon$: the residual standard deviation there; $[\tau_1, \tau_2]$: the days summed
  - AR: the return beyond what the market move predicts; $\sqrt{L}$: the SD of a sum of L independent residuals grows with $\sqrt{L}$
- For unanticipated news, efficiency predicts a jump on day 0 and no predictable drift or reversal afterwards
- Daily data can only show absorption within a day; speed measured in microseconds needs intraday data (Chapter 10); literature: ▸ Appendix

Worked Example: the Romanian Bank Tax of December 2018 (1/2)



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Worked Example: the Romanian Bank Tax of December 2018 (2/2)

- Tax announced on the evening of 18 Dec 2018, day 0 = 19 Dec; Banca Transilvania (TLV), BRD and their equal-weighted portfolio vs BET (Seminar 0, B7; Seminar 2, B9)
- TLV: $\hat{b} = 1.18$, $\hat{\sigma}_\varepsilon = 0.83\%$, $AR_0 = -8.3\%$ ($t = -10.0$); $CAR[0, +5] = -1.1\%$ ($t = -0.54$), since $CAR[+1, +5] = +7.2\%$ ($t = 3.82$)
- Most of the day-0 fall reverses within the week: an **overreaction**, against semi-strong efficiency, or later news about the final form of the ordinance; one event cannot separate them

MFM_ch2_seminar

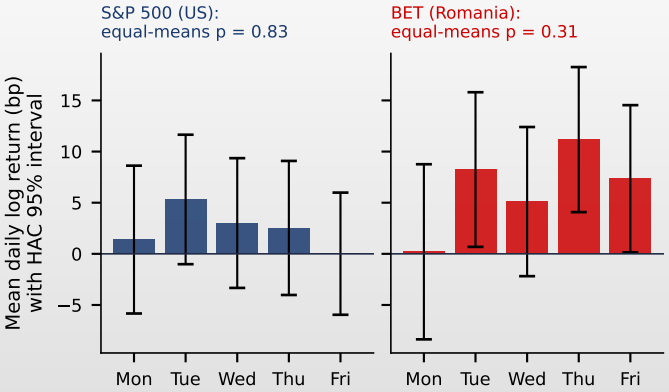
Event Studies: Inference Done Properly

- Prediction-error variance: $\mathrm{Var}(AR_\tau) = \sigma_\varepsilon^2\left[1 + \frac{1}{T_0} + \frac{(r_{m,\tau} - \hat{\mu}_m)^2}{(T_0 - 1)\hat{\sigma}_m^2}\right]$, T_0 = estimation days MacKinlay (1997)
  - $\hat{\mu}_m$, $\hat{\sigma}_m^2$: mean and variance of the market return in the estimation window; the bracket grows when the event-day market move is unusual
  - Bank tax, BET $r_{m,0} = -11.9\%$: the bracket is 2.12, so TLV's t shrinks in absolute value from -10.0 to -6.85; BRD: $AR_0 = -5.6\%$, $t = -4.03$
- Several firms: standardise each AR by its own prediction error and use the cross-sectional variance, robust to event-induced variance Boehmer, Musumeci & Poulsen (1991)
- Same event day for all firms: abnormal returns are cross-correlated and cross-sectional tests over-reject; the equal-weighted portfolio test already contains the correlation
  - TLV and BRD: $\bar{\varrho} = 0.153$; mean AR_0 with an independence $t = -7.55$, portfolio $t = -7.03$
  - $\bar{\varrho}$: the average correlation between the firms' abnormal returns; N: number of firms
  - Kolari & Pynnönen (2010) correct a cross-sectional test by $\sqrt{(1 - \bar{\varrho})/(1 + (N - 1)\bar{\varrho})}$; with ten such firms, $(1 + 9\bar{\varrho})/(1 - \bar{\varrho}) = 2.81$: the test would understate the variance of the mean AR 2.81 times

Anomalies and Calendar Effects

- An **anomaly** is a return pattern the asset-pricing model cannot explain: an omitted risk factor (misspecified model), mispricing (behavioural bias) or data mining
  - Families: calendar, time-series (index momentum), cross-sectional (size, value, momentum: **Chapter 3**)
  - Returns 26% lower out of sample and 58% lower after publication McLean & Pontiff (2016)
- Calendar effects: Monday French (1980); January Rozeff & Kinney (1976); turn of the month (TOM: last trading day and first three of the next month) Lakonishok & Smidt (1988); monthly effect (first half of the month) Ariel (1987)
  - Day of week: $r_t = \sum_{d=1}^{5} \beta_d D_{d,t} + u_t$, Wald test of $\beta_1 = \cdots = \beta_5$; TOM and January: one dummy each
  - $D_{d,t} = 1$ if day t is weekday d (Monday, ..., Friday), else 0; β_d: the mean return on weekday d; u_t: the residual
  - Newey–West HAC errors Newey & West (1987), 5 daily or 3 monthly lags: residuals are heteroskedastic and autocorrelated

Day-of-the-Week Effect: S&P 500 and BET



- Mean daily log return by weekday, in basis points (bp; 1 bp = 0.01%), 2000–2026, HAC 95% intervals
- S&P 500: from 0.0 bp (Friday) to 5.3 bp (Tuesday); equal-means $p = 0.83$: no Monday effect since 2000, unlike French (1980)
- BET: Thursday 11.2 bp ($t = 3.09$), yet the joint test gives $p = 0.31$
- Two different nulls: a day's mean equal to zero (mixes the day effect with the positive average return) vs all five means equal (the day effect itself)

MFM_ch2_calendar_anomalies

January and Turn-of-the-Month Effects

Market	January extra (%/month)	t_{HAC}	TOM extra (bp/day)	t_{HAC}	p
S&P 500 (US)	−0.66	−0.80	3.2	0.93	0.3539
BET (Romania)	3.48	1.49	17.3	3.62	0.0003
Euro Stoxx 50	−0.30	−0.29	5.1	1.21	0.2266
WIG20 (Poland)	0.53	0.40	8.2	1.73	0.0841
BUX (Hungary)	2.66	2.25	15.1	3.63	0.0003

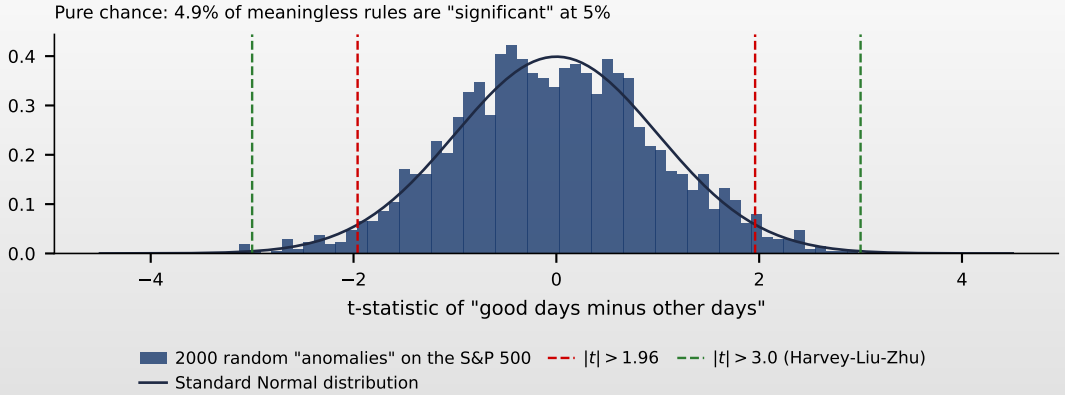
- TOM: BET +17.3 bp per day and BUX +15.1 bp, both with $|t| > 3.5$; S&P 500 only +3.2 bp
- January: large but imprecise for BET ($t = 1.49$); BUX, with $t = 2.25$, is the only market with a January-effect p-value below 0.05
- We ran 15 tests (5 markets $\times$ 3 effects): which survive a correction for multiple testing?

Multiple Testing: Bonferroni and Holm

- **FWER** (family-wise error rate): the probability of at least one false rejection; with M independent tests at level α, FWER $= 1 - (1-\alpha)^M$
  - $M = 15$, $\alpha = 5\%$: FWER $= 53.7\%$, a coin flip
- **Bonferroni**: reject only if $p_i < \alpha/M$; guarantees FWER $\leq \alpha$ for any dependence
- **Holm**: sort $p_{(1)} \leq \cdots \leq p_{(M)}$; reject while $p_{(j)} < \alpha/(M-j+1)$, stop at the first failure
  - Same guarantee, never less powerful than Bonferroni
- Our 15 calendar tests: Bonferroni threshold 0.0033; 3 raw rejections, 2 survive Holm (BET and BUX turn of the month); Holm stops at BUX January
- Seminar B4 carries out the ranking step by step; bootstrap methods that use the dependence between tests:

Appendix

How Many "Anomalies" Does Pure Chance Produce? (1/2)



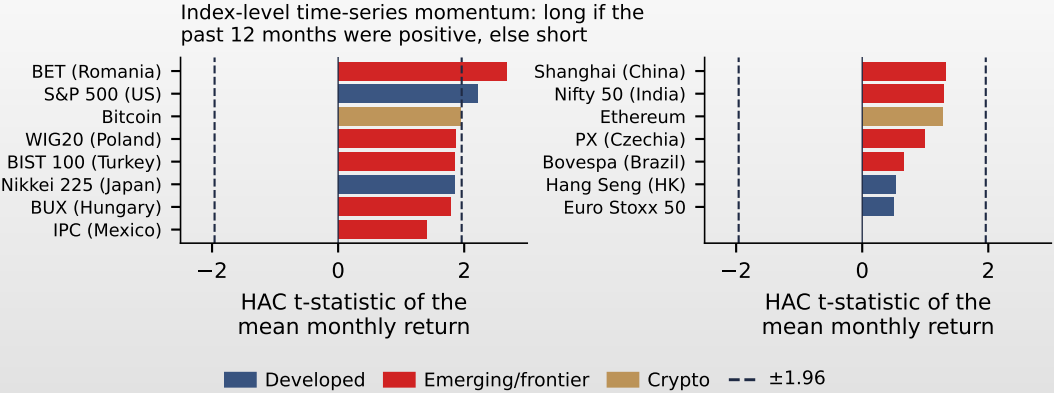
MFM_ch2_multiple_testing

How Many "Anomalies" Does Pure Chance Produce? (2/2)

- 2 000 meaningless rules on the S&P 500: each labels a random 20% of days as "good" and tests good minus other days
- 4.9% have $|t| > 1.96$, the 5% the test promises; the largest $|t|$ is 3.19
- Whoever tries many rules and reports the best is very likely to find a "significant anomaly"; data snooping in trading rules: Sullivan, Timmermann & White (1999)
- Harvey, Liu & Zhu (2016) suggest $|t| > 3.0$ for a new factor; here 0.25% of random rules pass. A higher bar is a heuristic, not a substitute for defining the family of tests tried

MFM_ch2_multiple_testing

Momentum and Index-Level Time-Series Momentum (1/2)



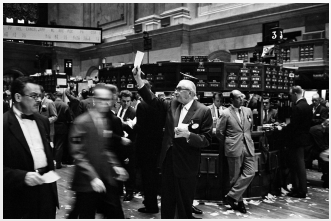
MFM_ch2_tsmom

Momentum and Index-Level Time-Series Momentum (2/2)

- Reversal of 3–5-year losers De Bondt & Thaler (1985); cross-sectional momentum over 3–12 months Jegadeesh & Titman (1993) (Chapter 3)
- TSMOM Moskowitz, Ooi & Pedersen (2012): signal $s_t = \text{sign}\left(\sum_{j=1}^{12} r_{t-j}\right)$
  - $+1$ (long) if the past 12 monthly log returns sum to a gain, -1 (short) otherwise
  - strategy return $s_t R_t$, R_t the index return of month t; bars: HAC t (6 lags)
- Significant: BET ($t = 2.68$, 1.40% a month), S&P 500 ($t = 2.22$); Bitcoin $t = 1.94$
- Sharpe ratio below buy-and-hold: BET 0.65 vs 0.75, S&P 500 0.48 vs 0.52: much of TSMOM is the equity premium earned while long
- Data limits: price indices without financing rates or dividends, 15 markets tested; full table: Appendix

MFM_ch2_tsmom

Psychology in Prices, from the Floor to the Feed



New York Stock Exchange trading floor, 1963

Photo: Thomas J. O'Halloran (1963); public domain; Wikimedia Commons

- **Excess volatility** Shiller (1981): prices move far more than the dividends that should justify them; **overreaction**: the source of the 3–5-year reversal De Bondt & Thaler (1985); portraits: ▸ Appendix
- **Prospect theory** Kahneman & Tversky (1979): losses loom larger than gains relative to a reference point
  - Disposition effect: winners sold too early, losers held too long
- **Overconfidence**: too much trading; **anchoring** and **limited attention**: slow reaction to news, a candidate cause of momentum
- **Herding** Pele, Mazurencu-Marinescu & Nijkamp (2013): once voices on a floor, now order books, algorithms and social media (Chapter 10)
- Biases alone do not break efficiency: rational arbitrageurs could trade against them. Can they?

Limits to Arbitrage



Andrei Shleifer, Harvard

Photo: Manuel Castells (2018); CC BY-SA 2.0; Wikimedia Commons

- Shleifer & Vishny (1997): real arbitrage is risky and needs capital
- **Noise-trader risk** De Long, Shleifer, Summers & Waldmann (1990): mispricing can get worse before it corrects
  - Noise traders: investors who trade on signals without information content (sentiment, rumours)
- Funding constraints: managers lose clients after losses and must close positions at the worst time
- Costs: short-selling limits, margins, trading costs, no perfect substitutes
- Anomalies can persist where arbitrage is hard: in small, illiquid markets such as BET we indeed find the strongest persistence
- The AMH reconciles both views: rational and biased behaviour coexist and evolve

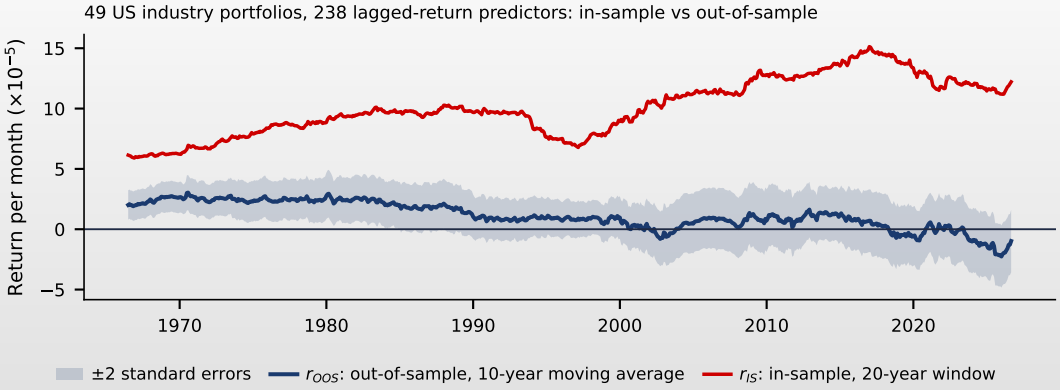
Can Machine Learning Beat Efficient Markets?

- ML (machine learning) learns a prediction rule from data and captures nonlinear patterns that autocorrelation tests miss (Chapter 13)
- Gu, Kelly & Xiu (2020): trees and neural networks give large economic gains over linear regressions in the US cross-section, yet their monthly out-of-sample R^2 is below 1%
- Large language models read news and filings: Tak & Pele (2025) forecast stock trends with LLaMA
- What ML adds to the cautions of this chapter: **chronological validation**
  - train only on data available before each forecast (walk-forward); random cross-validation leaks the future (look-ahead bias)
- Key question: when many predictors are available, what does an in-sample rejection of efficiency mean? The case study answers it

Case Study: Rational Learning with Many Predictors

- Martin & Nagel (2022): investors price N assets whose cash flows depend on J characteristics with **unknown** coefficients and learn them optimally (Bayes) from past data; research question: Appendix
  - the posterior mean is a ridge regression (OLS with a penalty on the squared coefficients): coefficients shrunk towards 0; investors are risk-neutral, so no predictability can be a risk premium
- **In sample**: with $J/N \to \psi > 0$ the test of no predictability rejects with probability tending to 1 (Martin & Nagel (2022), Prop. 4); formal results: Appendix
  - ψ: the limit of the number of predictors per asset, J/N
  - a rejection is not evidence of irrationality: the rational-expectations null is the wrong benchmark when parameters must be learned
- **Out of sample**: a portfolio built with coefficients from other periods has $\mathbb{E}[r_{OOS}] = 0$ (Martin & Nagel (2022), Prop. 6)
  - r_{OOS}: the return of that portfolio in a period not used to estimate its coefficients
  - out-of-sample tests keep their economic meaning: they are the right test of efficiency
- US stocks: in-sample R^2 about 1%, stable; out of sample much lower and within two standard errors of 0 after 2014 (Martin & Nagel (2022), Fig. 8). Model, simulations and design: Appendix

Replication on Course Data: In Sample vs Out of Sample (1/2)



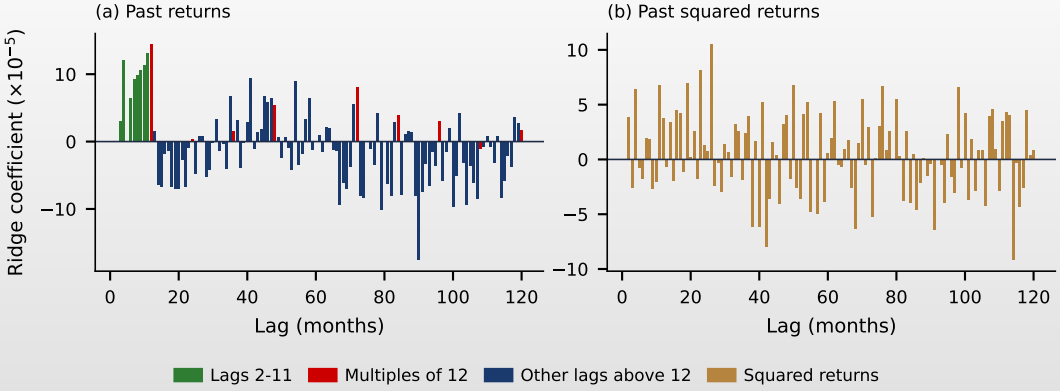
MFM_ch2_big_data

Replication on Course Data: In Sample vs Out of Sample (2/2)

- 49 industry portfolios (Kenneth French Data Library), July 1926 – August 2026, 238 lagged-return predictors, rolling 20-year OLS, specification of Martin & Nagel (2022), Sec. 7
- r_{IS} (in-sample return of the predicted-return portfolio): 5.90 to 15.14 ($\times 10^{-5}$ a month), positive by construction; mean in-sample $R^2 = 5.4\%$
- r_{OOS} (out-of-sample return of the same portfolio), 842 months: mean 1.05×10^{-5}, about 11% of r_{IS}; to June 1986: 4.16 SE above 0, afterwards 0.62
- Bands (± 2 SE) are descriptive, not a test of $\mathbb{E}[r_{OOS}] = 0$; 49 portfolios instead of thousands of stocks make J/N large; details: ▸ Appendix

MFM_ch2_big_data

Replication on Course Data: Ridge Coefficients by Lag (1/2)



MFM_ch2_big_data

Replication on Course Data: Ridge Coefficients by Lag (2/2)

- Full-sample ridge, January 1971 – August 2026, penalty by leave-one-year-out cross-validation (56 years); compare Martin & Nagel (2022), Fig. 7
- Lags 2–11: 8 of 10 positive, **momentum** Jegadeesh & Titman (1993); multiples of 12: 9 of 10 positive, **seasonality** Heston & Sadka (2008)
- Other lags above 12: 67% negative, long-run **reversal** De Bondt & Thaler (1985); squared returns: no dominant sign
- Cross-validated $R^2 = 0.013\%$ (OLS: -2.26%): shrinkage recovers the known anomalies, yet they predict almost nothing in the years left out; limits and extensions: Appendix

MFM_ch2_big_data

Hypothesis: Did Spot ETFs Make Bitcoin More Efficient?

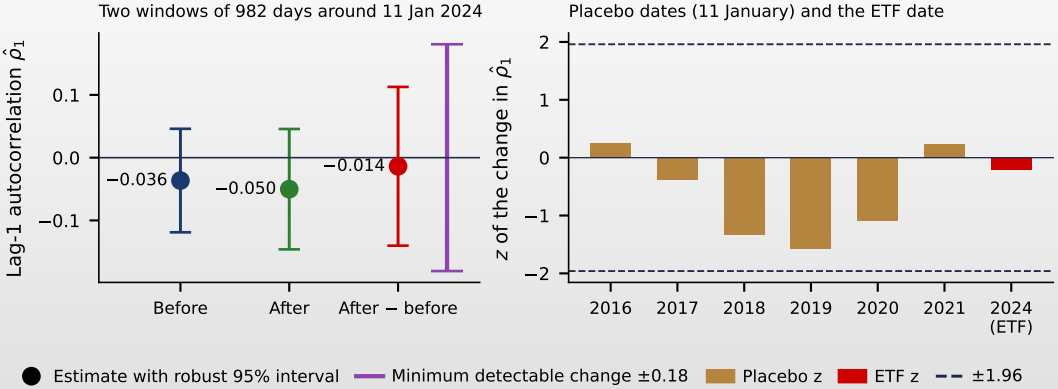
- A typical assistant proposal: "US spot Bitcoin ETFs (trading since 11 January 2024) brought arbitrageurs, so Bitcoin became more efficient"
- Discovery loop in one pass; reusable prompts: Appendix
  - **literature**: Kia, Liu, Li, Song & Xu (2025) find that after the spot ETFs the spot market became the dominant venue for price discovery again; Tran & Leirvik (2020): crypto efficiency varies and has increased; resolve every DOI yourself
  - **hypothesis**: $H_0 : \rho_{1,\text{post}} = \rho_{1,\text{pre}}$, two windows of equal length around the launch date (first trading day)
  - **data and code**: daily Bitcoin closing price (USD); first replicate the chapter's full-sample $\hat{\rho}_1 = -0.022$
- Checks fixed before seeing results; rationale for each: Appendix
  - robust standard errors (not $1/\sqrt{T}$); equal windows; the launch date only, plus placebo dates
  - a control asset for confounders (the April 2024 halving, rates, inflows)
  - a correction for every statistic tried; the minimum detectable effect

Estimation and Power

- Test: two windows of 982 days around 11 January 2024; $z = \Delta\hat{\rho}_1 / \sqrt{s_{\text{pre}}^2 + s_{\text{post}}^2}$, robust standard errors s
  - $\Delta\hat{\rho}_1 = \hat{\rho}_{1,\text{post}} - \hat{\rho}_{1,\text{pre}}$; s_{pre}, s_{post}: the robust SEs of the two estimates, independent windows
  - before (May 2021 – Jan 2024): $\hat{\rho}_1 = -0.036$ ($s = 0.042$), VR(5) $= 1.000$ ($z^* = 0.00$)
  - after (Jan 2024 – Sep 2026): $\hat{\rho}_1 = -0.050$ ($s = 0.049$), VR(5) $= 0.939$ ($z^* = -0.64$)
  - $\Delta\hat{\rho}_1 = -0.014$, $z = -0.21$: an unchanged ρ_1 is not rejected
- **Minimum detectable effect (MDE)**: the smallest true change detected with 80% power at the 5% level
  - MDE $= (z_{0.975} + z_{0.80}) \times \text{SE}(\Delta\hat{\rho}_1) \approx 2.8 \times \text{SE} = 0.18$, z_p: the p-quantile of the standard Normal distribution (1.96 and 0.84)
  - if Bitcoin was already close to efficient before 2024, this design could not detect a smaller improvement
- Placebo at 11 January 2016–2021: 0 of 6 significant, largest $|z| = 1.57$
- Causality would also need a control asset (Ethereum before its own ETFs in July 2024), parallel pre-trends, no anticipation and no spillovers

ai_discovery_case.py

Mini-Case: Estimate, Placebo and MDE (1/2)



ai_discovery_case.py

Mini-Case: Estimate, Placebo and MDE (2/2)

- Left: $\hat{\rho}_1$ before and after, with robust 95% intervals; the difference with $\pm$MDE
- Right: the ETF z lies inside the range produced by six placebo dates
- "Not significant" is not "no effect": the MDE is several times the estimate
- A better outcome: intraday price discovery across venues, not daily ρ_1

ai_discovery_case.py

Your Turn: A Project Seed

- **Question**: did the spot ETFs change the time-varying efficiency of Bitcoin, relative to a control asset?
  - rolling VR and DFA (this chapter) for Bitcoin and Ethereum; Ethereum is untreated until July 2024
  - if parallel trends, no anticipation and no spillovers are not credible, report descriptive, not causal, differences
- **Replicate first**: the chapter's yearly Bitcoin autocorrelations and the mini-case numbers
- **Pre-register**: H_0, H_1, dates, windows and corrections, committed before the results
- **Report**: robust inference, placebo tests and the MDE; the code as a Quantlet
- **Log**: prompts in `AI_USE.md`; every caught error (an i.i.d. standard error, an invented reference) in `AI_ERRORS.md`

Synthesis, Check Yourself, and Next: Chapter 3

Five steps for every efficiency claim

- **Information set**: which $\mathcal{I}_t$ (past prices, public news)?
- **Null**: RW1, RW3 or the martingale difference, stated together with the risk model
- **Statistic**: its variance valid under volatility clustering
- **Uncertainty**: horizons, markets and windows tried; out-of-sample evidence
- **Economics**: rejection $\neq$ exploitable profit $\neq$ irrational market

Check yourself

- Why can RW3 be rejected for the S&P 500 while the market remains efficient in the economic sense?
- Why does BET's $H(20) > 0.5$ not prove long memory?
- Which of the two results, BET's persistence or Bitcoin's non-rejection, would survive a change of test, window and inference?
- Next, Chapter 3: risk-adjusted returns with factor models, and the factor zoo
- Advanced courses and textbooks: Appendix

Appendix: Pioneers, Bachelier and Samuelson



Louis Bachelier (1870–1946)

Photo: unknown author; public domain; Wikimedia Commons



Paul A. Samuelson

Photo: Bernard Gotfryd (1970–1975); public domain; Wikimedia Commons

- Bachelier (1900): prices of French government bonds move like a random walk
  - "The mathematical expectation of the speculator is zero"
- Samuelson (1965): if prices are **properly anticipated**, their changes are unpredictable
- Both results are the ancestors of the Efficient Market Hypothesis

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Appendix: Eugene Fama



Eugene F. Fama, Nobel Prize 2013

Photo: Bengt Nyman (2013); CC BY 2.0; Wikimedia Commons

- Fama (1970) gave the definition used in this chapter and the three forms of efficiency
- Nobel Prize 2013, shared with Robert Shiller (a sceptic of efficiency) and Lars Peter Hansen
- Update and survey: Fama (1991); accessible critique: Malkiel (2003)

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Appendix: Summary Table, Weak-Form Tests 2000–2026

Market	T	$\hat{\rho}_1$	robust SE	$\widehat{\mathrm{VR}}(5)$	$z^*(5)$	CD p	Runs z
S&P 500 (US)	6 714	−0.099	0.027	0.830	−2.77	0.001	4.34
Euro Stoxx 50	6 836	−0.029	0.019	0.910	−2.08	0.097	3.41
Nikkei 225 (Japan)	6 545	−0.038	0.024	0.923	−1.48	0.317	1.97
Hang Seng (HK)	6 583	−0.004	0.024	0.967	−0.64	0.872	−0.17
BET (Romania)	6 670	0.109	0.028	1.173	2.98	0.001	−6.05
WIG20 (Poland)	6 685	0.018	0.018	1.009	0.24	0.777	2.25
BUX (Hungary)	6 666	0.037	0.024	1.013	0.26	0.427	−0.42
PX (Czechia)	6 682	0.052	0.032	1.041	0.59	0.350	−2.53
BIST 100 (Turkey)	6 692	0.005	0.024	1.033	0.64	0.949	−0.63
Bovespa (Brazil)	6 622	−0.028	0.025	0.928	−1.29	0.521	0.88
IPC (Mexico)	6 689	0.070	0.019	1.039	0.93	0.001	−3.27
Nifty 50 (India)	6 606	0.048	0.024	1.048	0.91	0.178	−5.46
Shanghai (China)	6 474	0.021	0.019	1.046	1.11	0.409	1.02
Bitcoin	4 384	−0.022	0.026	0.991	−0.19	0.835	2.68
Ethereum	3 913	−0.012	0.027	1.067	1.17	0.027	3.91
EUR/RON (reference rate)	5 352	0.170	0.036	1.157	2.19	0.000	−1.66

- T: number of daily returns; CD: Chow–Denning over $q = 2, 5, 10, 20$; EUR/RON from July 2005, crypto on 365 days per year

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Appendix: Runs Test on All Markets, Other VR Tests

- Runs z (table on the previous slide): Nifty 50 -5.46, IPC -3.27 (persistence in emerging markets); Ethereum 3.91, significant although $\hat{\rho}_1$ is not
- Choi (1999): an automatic VR test that chooses the horizon from the data (spectral estimate at frequency zero)
- Wild-bootstrap versions for small samples: Kim (2006); applications to Asian markets: Kim & Shamsuddin (2008)

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Appendix: Automatic Portmanteau Test of the MDH

- Escanciano & Lobato (2009): $AQ = T\sum_{j\le\tilde{p}}\hat{\gamma}_j^2/\hat{\kappa}_j \to \chi^2_1$, with $\hat{\gamma}_j = \frac{1}{T-j}\sum_t e_t e_{t-j}$ and $\hat{\kappa}_j = \frac{1}{T-j}\sum_t e_t^2 e_{t-j}^2$ (robust to conditional heteroskedasticity)
  - $\tilde{p} \le 10$ maximises $Q_p = T\sum_{j\le p}\hat{\gamma}_j^2/\hat{\kappa}_j$ minus a penalty that switches from BIC ($p \ln T$) to AIC ($2p$) when some $\sqrt{T}|\hat{\rho}_j|$ is large
- $e_t = r_t - \bar{r}$; $\hat{\gamma}_j$: autocovariance at lag j; $\hat{\kappa}_j$: its robust variance; a large AQ rejects the MDH
- AQ: S&P 500 13.1 ($p = 0.0003$), BET 14.9 ($p = 0.0001$), Bitcoin 0.8 ($p = 0.385$); the MDH is rejected at 5% for 5 of the 16 series

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Appendix: Predictive Regression, Data and Full Output

- Monthly log excess return (dividends included, minus the 3-month T-bill rate) regressed on the lagged dp_t
  - Data: Shiller (2026) (S&P Composite, to June 2026), extended with our S&P 500 closes and the dividends of the SPY fund; 3-month US Treasury bill yield (FRED, series TB3MS; annual %), monthly log rate $\ln(1 + \text{TB3MS}/1200)$ lagged one month
- OLS: $\hat{\beta} = 0.267$, Newey–West SE 0.293 (12 lags), $t = 0.91$, OLS $t = 1.16$, $R^2 = 0.12\%$
- Corrected slope / unadjusted HAC SE $= -0.22$: only illustrative, it ignores the uncertainty of the correction
- Economic value Campbell & Thompson (2008) (number: Out-of-Sample Evidence slide): proportional rise in the expected return of the market timer
  - $\dfrac{R^2}{1-R^2} \cdot \dfrac{1+S^2}{S^2}$, S the monthly Sharpe ratio
- Diebold–Mariano Diebold & Mariano (1995) $t = -0.57$ for the same forecasts, against Clark–West $t = 0.42$

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Appendix: Harold Hurst and the Nile



Harold Edwin Hurst (1880–1978)

Photo: Elliott & Fry (1953); public domain; Wikimedia Commons



Nilometer on Elephantine Island, Aswan

Photo: Olaf Tausch (2011); CC BY 3.0; Wikimedia Commons

- Hurst (1951) designed Nile reservoirs; his finding and the **rescaled range**: R/S slide

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Appendix: The Anis–Lloyd Expectation

- Anis & Lloyd (1976): exact $\mathbb{E}[R_n/s_n]$ under i.i.d. returns from the Normal distribution
  - $\mathbb{E}[R_n/s_n] = \dfrac{\Gamma((n-1)/2)}{\sqrt{\pi}\,\Gamma(n/2)} \sum_{i=1}^{n-1} \sqrt{\dfrac{n-i}{i}}$ (Γ: the gamma function)
- This $\mathbb{E}[R_n/s_n]$ is the reference subtracted in H_{AL} (R/S Analysis slide)
- Uncorrected estimates overstate memory by about 0.04–0.05 with daily data

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Appendix: Long Memory with Inference, the Local Whittle Estimator

- $d = H - 0.5$; spectral density $f(\lambda) \sim G\lambda^{-2d}$ as $\lambda \to 0$ ($G > 0$ a constant; $d > 0$: the density explodes at low frequencies, long memory); fit only the first m Fourier frequencies $\lambda_j = 2\pi j/T$
  - $\hat{d} = \arg\min_d \left[\ln\left(\frac{1}{m} \sum_{j=1}^{m} \lambda_j^{2d} I(\lambda_j) \right) - \frac{2d}{m} \sum_{j=1}^{m} \ln \lambda_j \right]$, I: the periodogram
  - Robinson (1995): for $|d| < 1/2$, $m \to \infty$, $m/T \to 0$, a smooth f near 0 and a linear process with conditionally homoskedastic martingale-difference innovations, $\sqrt{m}(\hat{d} - d) \to N(0, 1/4)$ without Gaussianity: a confidence interval instead of a band
- Estimates with $m = T^{0.65}$ and the 95% CI $\hat{d} \pm 1.96/(2\sqrt{m})$
  - S&P 500 -0.013 $[-0.069, 0.043]$; Bitcoin 0.063 $[-0.002, 0.127]$
  - BET 0.118 $[0.062, 0.174]$; Ethereum 0.112 $[0.045, 0.179]$
- Bandwidth trade-off: a large m reaches frequencies where short-run autocorrelation (BET $\hat{\rho}_1 = 0.109$) biases $\hat{d}$ upward
  - with $m = T^{0.5}$ BET still gives 0.135 (SE 0.056): a sensitivity check supporting, not proving, that its $d > 0$ is not only short memory
  - GARCH returns also strain the conditions above

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Appendix: EMH versus AMH

	EMH	AMH
Investors	Rational, optimising	Bounded rationality, heuristics
Efficiency	An equilibrium property, tested as a restriction	A matter of degree that changes with conditions
Risk–return	Equilibrium risk premia, possibly time-varying	Changes with the population of investors
Anomalies	Omitted risk factors or data mining	Transient profit opportunities
Empirical tool	Tests of an equilibrium restriction	Tests of time and state variation (rolling windows)

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Appendix: Is Bitcoin Efficient? The Literature

- Urquhart (2016): Bitcoin inefficient over 2010–2016; some tests indicate efficiency in the later subperiod
  - His battery (Ljung–Box, runs, the rank test of Bartels (1982), automatic VR, BDS, R/S) includes BDS, which rejects under volatility clustering alone: not a test of RW3
- Nadarajah & Chu (2017): with a power transformation of returns, Bitcoin is weak-form efficient
- Kristoufek (2018): Bitcoin markets mostly inefficient in 2010–2017, with efficiency changing over time
- Tran & Leirvik (2020): time-varying efficiency across many crypto-assets, increasing over time
- Pele et al. (2023): crypto-assets have become a distinct asset class; their returns share the stylised facts of other markets
- Year by year in this chapter: Bitcoin 2011–2013 $\hat{\rho}_1 = -0.002, -0.071, 0.069$; the largest $|z^*(5)|$ in any year is 1.71

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Appendix: Romania, Frontier then Emerging

- FTSE Russell (2019): Romania reclassified from Frontier to Secondary Emerging, announced in September 2019, effective 21 September 2020
- Expected channels: more foreign investors, more analysts, more liquidity $\Rightarrow$ faster price discovery
- Earlier BVB evidence: crashes and log-periodic bubbles Pele & Mazurencu-Marinescu (2012); herding in an illiquid market Pele, Mazurencu-Marinescu & Nijkamp (2013)
- Did efficiency improve? Seminar C2 compares five years before and after, with WIG20 and BUX as a comparison group and a block bootstrap
  - Block bootstrap: resamples blocks of consecutive days to keep the dependence Künsch (1989); a before/after comparison is descriptive unless confounders are controlled

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Appendix: How Fast Is "Fast"?

- At high frequency, futures prices react to scheduled macro news within microseconds; Tak, Schlamp, Osterrieder & Pele (2026) measure this race at Eurex and CME
- Crypto events: Pele, Lin, Wang & Gaman (2025) study market responses to Ethereum development milestones
- Speed of adjustment is itself a measure of efficiency, and it depends on technology (Chapter 10, market microstructure)

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Appendix: Dependent Tests, Reality Check, SPA and StepM

- Bonferroni and Holm ignore the correlation between tests (same markets, same days)
  - White (2000): Reality Check, bootstrap of $\max_j$ over all rules tried; Hansen (2005): SPA, studentised and less sensitive to poor rules
  - Romano & Wolf (2005): StepM (Romano–Wolf step-down procedure), step-down control of the FWER from the **joint** bootstrap distribution of all statistics
- Our 15 calendar tests: StepM with a bootstrap of blocks of 3 calendar months, common to all markets, 999 replications
  - Critical p at the first step 0.0016 (Bonferroni: 0.0033); 2 rejections, BET and BUX TOM, as with Holm
  - Why stricter than Bonferroni? At nominal 5%, the bootstrap rejection rates of the HAC tests are 6.0% (day of week), 6.4% (TOM), 6.4% (January)
- From over two million simulated strategies: thresholds $|t| > 3.8$ (time series) and 3.4 (cross-section); ignoring them makes about 45% of rejections false Chordia, Goyal & Saretto (2020); lowering false discoveries raises missed ones Harvey & Liu (2020)

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Appendix: Time-Series Momentum, All Markets

Market	Mean (%/month)	t_{HAC}	Sharpe TSMOM	Sharpe buy-and-hold	$\hat{\rho}_1$ monthly
S&P 500 (US)	0.61	2.22	0.48	0.52	0.03
Euro Stoxx 50	0.16	0.51	0.11	0.16	0.05
Nikkei 225 (Japan)	0.61	1.85	0.39	0.42	0.09
Hang Seng (HK)	0.18	0.53	0.10	0.19	0.04
BET (Romania)	1.40	2.68	0.65	0.75	0.19
WIG20 (Poland)	0.67	1.86	0.36	0.26	0.02
BUX (Hungary)	0.72	1.79	0.40	0.64	0.10
PX (Czechia)	0.38	0.99	0.24	0.45	0.10
BIST 100 (Turkey)	1.03	1.85	0.37	0.75	−0.08
Bovespa (Brazil)	0.28	0.65	0.14	0.51	0.11
IPC (Mexico)	0.44	1.41	0.32	0.63	0.03
Nifty 50 (India)	0.50	1.29	0.28	0.64	0.02
Shanghai (China)	0.59	1.32	0.30	0.23	0.08
Bitcoin	4.47	1.94	0.72	1.08	0.13
Ethereum	6.50	1.28	0.57	0.90	0.16

- 307 months for equities, fewer for crypto; Sharpe ratio $\sqrt{12}\,\bar{x}/s_x$ of price returns ($\bar{x}$, s_x: mean and SD of the monthly strategy returns), no risk-free rate subtracted; Moskowitz, Ooi & Pedersen (2012) use futures excess returns scaled to 40% ex-ante volatility

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Appendix: Robert Shiller and Richard Thaler



Robert J. Shiller, Nobel Prize 2013

Photo: Bengt Nyman (2013); CC BY 2.0; Wikimedia Commons



Richard H. Thaler, Nobel Prize 2017

Photo: Bengt Nyman (2017); CC BY 2.0; Wikimedia Commons

- Excess volatility Shiller (1981) and overreaction De Bondt & Thaler (1985): Psychology in Prices slide

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Appendix: Case Study, Research Question

- Martin & Nagel (2022): investors price N assets whose cash flows depend on J characteristics with unknown coefficients; what does a rejected no-predictability test mean when J is comparable to N?
  - the coefficients must be learned from past data: a high-dimensional prediction problem, as in machine learning
- Semi-strong efficiency: prices "fully reflect" public information Fama (1970); two readings
  - investors know the parameters of the cash-flow model (rational expectations: the null of most efficiency tests)
  - investors update their beliefs optimally (Bayes) with the data available in real time
- Why it matters: hundreds of in-sample return predictors, the "factor zoo" Harvey, Liu & Zhu (2016); text, satellite images and social media keep increasing J
  - does every significant predictor need a risk-based or a behavioural explanation?
- Link to this chapter: in-sample vs out-of-sample predictability Campbell & Thompson (2008), Goyal, Welch & Zafirov (2024), and the decay of anomalies after publication McLean & Pontiff (2016)

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Appendix: Case Study, Model and Method (1/2)

- **Model** (Martin & Nagel (2022), Sec. 2)
  - cash-flow growth $\Delta y_t = Xg + e_t$, with $e_t \sim N(0, I)$ and a zero interest rate
  - Δy_t: the N cash-flow growth rates; X: the $N \times J$ matrix of characteristics; g: the J unknown coefficients; I: identity matrix
- **Learning** (Prop. 1–2)
  - prior $g \sim N(0, (\theta/J)\, I)$, the true distribution from which g is drawn; θ: signal strength
  - realised return = underreaction caused by shrinkage + estimation noise + unpredictable shock
- **Asymptotics and simulations** (Sec. 4)
  - $N, J \to \infty$ with $J/N \to \psi > 0$
  - simulations with $N = 1000$, entries of X drawn from the standard Normal distribution and $\theta = 1$

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Appendix: Case Study, Model and Method (2/2)

- **Tests**
  - in sample: a cross-sectional OLS of r_{t+1} on X (requires $J < N$) with a joint χ^2 test
  - out of sample: a portfolio with weights $\frac{1}{N} X h_{s+1}$, from coefficients estimated in other periods
  - $r_{\text{OOS},t+1} = \frac{1}{N} h'_{s+1} X' r_{t+1}$, $s \neq t$ (eq. 19); r_{IS}: the same with $s = t$
  - h_{s+1}: the estimated coefficient vector; r_{t+1}: the N returns
- **Empirical design** (Sec. 7)
  - US stocks in CRSP, without those below the 20th NYSE size percentile or priced below \$1
  - predictors: simple and squared monthly returns at lags $t-2$ to $t-120$ (238 predictors), demeaned and standardised across stocks every month
  - full-sample ridge, 1971 – June 2019, penalty chosen by leave-one-year-out cross-validation
  - rolling 20-year OLS gives r_{OOS} from January 1956

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Appendix: Case Study, Formal Results

- In sample (Prop. 4 of Martin & Nagel (2022), main slide): for $\psi > 0.4$ ($\theta = 1$, one period of learning) the probability of no rejection is of the order $e^{-0.015N}$ (Fig. 2)
- Simulations ($N = 1000$, four periods of learning): a test of nominal size 5% rejects with probability above 90% once $J > N/2$ (Fig. 3b)
  - not a size distortion: the in-sample predictable component really is in the prices
- Single-predictor tests with k-FWER control (probability of k or more false rejections at most 5%): for $k = 20$ and $J/N = 0.05$, the probability of at least k rejections is 25% instead of 5%; about 80% for J/N close to 1 (Fig. 4)
  - multiple-testing corrections do not help: the problem is the null hypothesis, not the test
- Out of sample: $\mathbb{E}[r_{\text{OOS},t+1}] = 0$ holds forward and backward in time (Prop. 6); positive only if investors shrink more than the data warrant (Fig. 6)
- US stocks: the ridge coefficients recover momentum, long-run reversal and momentum seasonality (Fig. 7); $\eta_S \approx 0.006\%$ per month (Fig. 8)

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Appendix: Case Study, Replication, Limitations and Extensions

- **Replicate with the course data** (stock-level CRSP files are not part of them)
  - the 49 value-weighted industry portfolios of the main replication slides (Seminar 2, C3); same predictors, standardisation and windows
  - the size and price filters do not apply to portfolios
- **Limits stated by the authors**
  - lagged returns are equilibrium outcomes, not the exogenous X of the model: the application illustrates the wedge, it does not test the model
  - a t-test on the monthly r_{OOS} is not a valid test of $\mathbb{E}[r_{OOS}] = 0$: the expectation also averages over g
  - the backward result may rely on i.i.d. cash-flow growth; homogeneous investors and constant characteristics
- **Open extensions**
  - time series, as the authors suggest: does the dividend–price regression of this chapter show the same in-sample/out-of-sample wedge?
  - nonlinear models (trees, neural networks Gu, Kelly & Xiu (2020)) add functional flexibility without necessarily changing J: does the wedge grow?
  - AMH: is early out-of-sample predictability the excess shrinkage of investors with less computing power?

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Appendix: Case Study, In Sample vs Out of Sample in Detail

- $r_{\text{IS}} \geq 0$ in every 20-year window by construction: $r_{\text{IS}} = b' G^{-1} b/240$, with $G = \sum_s X_s' X_s / N_s$ and $b = \sum_s X_s' r_s / N_s$ over the 240 months s of the window
  - X_s: the $N_s \times 238$ predictor matrix of month s; r_s: the N_s portfolio returns; $G^{-1}b$: the pooled OLS coefficients
  - mean in-sample R^2 of 5.4%, against about 1% for individual stocks in Martin & Nagel (2022): 238 predictors against only 40–49 portfolios a month, pooled over 240 months, inflate the in-sample fit
- Out of sample (from July 1956), until June 1986: mean $r_{\text{OOS}} = 1.98 \times 10^{-5}$, 4.16 standard errors above 0; afterwards 0.36×10^{-5}, only 0.62 standard errors
  - the 10-year average is last above two standard errors in the window ending in December 1987; in the windows ending in 2014–2019 its largest ratio to the standard error is 1.25, as for stocks
- Interpretation suggested by Martin & Nagel (2022): the gap is consistent with investor learning, but does not establish it or rule out risk premia; early out-of-sample gains are also consistent with excess shrinkage before the computer era (Fig. 6)

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Appendix: The Discovery Loop, Reusable Prompts (1/2)

- **Literature**
  - `List peer-reviewed studies since 2020 on how ETF introductions change the efficiency or price discovery of the underlying. For each: DOI, design (event study, difference-in-differences (DiD), structural break), efficiency measure. Flag any entry you are unsure exists.`
  - Resolve every DOI on Crossref; the title and authors must match
- **Hypothesis**
  - `Give three mechanisms by which spot ETFs could change Bitcoin's serial dependence (arbitrage, investor mix, trading hours). For each, the sign of the change in rho1 and VR(q), and a variable that separates it from the others.`
- **Data and code**
  - `Read the daily Bitcoin closing prices from the course data; log returns; rho1 with a heteroskedasticity-robust standard error and the Lo-MacKinlay VR(5) with z*, in two windows of equal length around 11 January 2024.`

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Appendix: The Discovery Loop, Reusable Prompts (2/2)

- **Robustness**
  - `Write a pre-registered robustness plan: window lengths, weekday-only closes, VR(q) for q = 2 to 20 with Chow-Denning, placebo dates, and Ethereum as a control until its own spot ETFs (July 2024).`
- **Critique**
  - `Act as a hostile referee: if Bitcoin was already efficient before 2024, what could this test ever detect? Compute the minimum detectable change in rho1 and name a better outcome variable.`
- **Report**
  - Estimates, robust standard errors, the placebo distribution, every test run (with a multiple-testing correction)
  - Prompts that shaped a decision go into `AI_USE.md`
- The assistant drafts the tests; the human decides which one answers the question

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Appendix: What the Researcher Must Check

- **i.i.d. standard errors**: $1/\sqrt{T}$ rejects too often under volatility clustering
  - test: robust SE and z^*, as in the weak-form tests of this chapter
- **Unequal windows**: different power before and after; test: equal lengths, fixed in advance
- **Event date chosen from the data**
  - test: the launch date (first trading day) only, plus a placebo distribution at dates without an event
- **Confounding**: before/after is not causal; test: a control asset (Ethereum before July 2024) in a difference-in-differences
  - causal only under the conditions of the Estimation and Power slide; the common post-period (Jan–Jul 2024) is short
- **Many statistics**: ρ_1, VR(2–20), runs; test: Chow–Denning or Holm, not the smallest single p-value
- **Low power**: report the MDE of the Estimation and Power slide

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Appendix: Further Study on Quantinar

Advanced courses on quantinar.com

- Efficiency of cryptocurrency markets - A GMM-based analysis
- Generalized Method of Moments

Code, data and textbooks

- All charts and tables: github.com/danpele/MFM/tree/main/Quantlets/Ch_02
- Textbooks: Campbell, Lo & MacKinlay (1997), Ch. 2; Franke, Härdle & Hafner (2019)

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References (1/7)

- Andrews, D.W.K. (1991). Heteroskedasticity and autocorrelation consistent covariance matrix estimation. *Econometrica*, 59(3), 817–858.
- Anis, A.A., Lloyd, E.H. (1976). The expected value of the adjusted rescaled Hurst range of independent normal summands. *Biometrika*, 63(1), 111–116.
- Ariel, R.A. (1987). A monthly effect in stock returns. *Journal of Financial Economics*, 18(1), 161–174.
- Bachelier, L. (1900). Théorie de la spéculation. *Annales scientifiques de l'École Normale Supérieure*, 17, 21–86.
- Bartels, R. (1982). The rank version of von Neumann's ratio test for randomness. *Journal of the American Statistical Association*, 77(377), 40–46.
- Boehmer, E., Musumeci, J., Poulsen, A.B. (1991). Event-study methodology under conditions of event-induced variance. *Journal of Financial Economics*, 30(2), 253–272.
- Brock, W.A., Dechert, W.D., Scheinkman, J.A., LeBaron, B. (1996). A test for independence based on the correlation dimension. *Econometric Reviews*, 15(3), 197–235.
- Campbell, J.Y., Lo, A.W., MacKinlay, A.C. (1997). The Econometrics of Financial Markets. Princeton University Press.
- Campbell, J.Y., Thompson, S.B. (2008). Predicting excess stock returns out of sample: can anything beat the historical average?. *Review of Financial Studies*, 21(4), 1509–1531.
- Campbell, J.Y., Yogo, M. (2006). Efficient tests of stock return predictability. *Journal of Financial Economics*, 81(1), 27–60.
- Choi, I. (1999). Testing the random walk hypothesis for real exchange rates. *Journal of Applied Econometrics*, 14(3), 293–308.
- Chordia, T., Goyal, A., Saretto, A. (2020). Anomalies and false rejections. *Review of Financial Studies*, 33(5), 2134–2179.

References (2/7)

- Chow, K.V., Denning, K.C. (1993). A simple multiple variance ratio test. *Journal of Econometrics*, 58(3), 385–401.
- Clark, T.E., West, K.D. (2007). Approximately normal tests for equal predictive accuracy in nested models. *Journal of Econometrics*, 138(1), 291–311.
- De Bondt, W.F.M., Thaler, R. (1985). Does the stock market overreact?. *Journal of Finance*, 40(3), 793–805.
- De Long, J.B., Shleifer, A., Summers, L.H., Waldmann, R.J. (1990). Noise trader risk in financial markets. *Journal of Political Economy*, 98(4), 703–738.
- Diebold, F.X., Mariano, R.S. (1995). Comparing predictive accuracy. *Journal of Business & Economic Statistics*, 13(3), 253–263.
- Escanciano, J.C., Lobato, I.N. (2009). An automatic Portmanteau test for serial correlation. *Journal of Econometrics*, 151(2), 140–149.
- Escanciano, J.C., Velasco, C. (2006). Generalized spectral tests for the martingale difference hypothesis. *Journal of Econometrics*, 134(1), 151–185.
- Fama, E.F. (1970). Efficient capital markets: a review of theory and empirical work. *Journal of Finance*, 25(2), 383–417.
- Fama, E.F. (1991). Efficient capital markets: II. *Journal of Finance*, 46(5), 1575–1617.
- Fama, E.F., Fisher, L., Jensen, M.C., Roll, R. (1969). The adjustment of stock prices to new information. *International Economic Review*, 10(1), 1–21.
- Franke, J., Härdle, W.K., Hafner, C.M. (2019). Statistics of Financial Markets: An Introduction, 5th ed. Springer.
- French, K.R. (1980). Stock returns and the weekend effect. *Journal of Financial Economics*, 8(1), 55–69.

References (3/7)

- French, K.R. (2026). Kenneth R. French Data Library. Tuck School of Business, Dartmouth College.
- FTSE Russell (2019). FTSE Russell announces results of country classification review (press release, 26 September 2019).
- Goyal, A., Welch, I., Zafirov, A. (2024). A comprehensive 2022 look at the empirical performance of equity premium prediction. *Review of Financial Studies*, 37(11), 3490–3557.
- Grossman, S.J., Stiglitz, J.E. (1980). On the impossibility of informationally efficient markets. *American Economic Review*, 70(3), 393–408.
- Gu, S., Kelly, B., Xiu, D. (2020). Empirical asset pricing via machine learning. *Review of Financial Studies*, 33(5), 2223–2273.
- Hansen, P.R. (2005). A test for superior predictive ability. *Journal of Business & Economic Statistics*, 23(4), 365–380.
- Harvey, C.R., Liu, Y. (2020). False (and missed) discoveries in financial economics. *Journal of Finance*, 75(5), 2503–2553.
- Harvey, C.R., Liu, Y., Zhu, H. (2016). . . . and the cross-section of expected returns. *Review of Financial Studies*, 29(1), 5–68.
- Heston, S.L., Sadka, R. (2008). Seasonality in the cross-section of stock returns. *Journal of Financial Economics*, 87(2), 418–445.
- Hurst, H.E. (1951). Long-term storage capacity of reservoirs. *Transactions of the American Society of Civil Engineers*, 116(1), 770–799.
- Jegadeesh, N., Titman, S. (1993). Returns to buying winners and selling losers: implications for stock market efficiency. *Journal of Finance*, 48(1), 65–91.
- Kahneman, D., Tversky, A. (1979). Prospect theory: an analysis of decision under risk. *Econometrica*, 47(2), 263–291.

References (4/7)

- Kendall, M.G. (1954). Note on bias in the estimation of autocorrelation. *Biometrika*, 41(3–4), 403–404.
- Kia, K., Liu, B., Li, Q., Song, V., Xu, K. (2025). Price discovery in Bitcoin ETF market. *Financial Review*, 61(2), 435–449.
- Kim, J.H. (2006). Wild bootstrapping variance ratio tests. *Economics Letters*, 92(1), 38–43.
- Kim, J.H., Shamsuddin, A. (2008). Are Asian stock markets efficient? Evidence from new multiple variance ratio tests. *Journal of Empirical Finance*, 15(3), 518–532.
- Kim, J.H., Shamsuddin, A., Lim, K.-P. (2011). Stock return predictability and the adaptive markets hypothesis: evidence from century-long U.S. data. *Journal of Empirical Finance*, 18(5), 868–879.
- Kolari, J.W., Pynnönen, S. (2010). Event study testing with cross-sectional correlation of abnormal returns. *Review of Financial Studies*, 23(11), 3996–4025.
- Kristoufek, L. (2018). On Bitcoin markets (in)efficiency and its evolution. *Physica A*, 503, 257–262.
- Künsch, H.R. (1989). The jackknife and the bootstrap for general stationary observations. *Annals of Statistics*, 17(3), 1217–1241.
- Lakonishok, J., Smidt, S. (1988). Are seasonal anomalies real? A ninety-year perspective. *Review of Financial Studies*, 1(4), 403–425.
- Lim, K.-P., Brooks, R. (2011). The evolution of stock market efficiency over time: a survey of the empirical literature. *Journal of Economic Surveys*, 25(1), 69–108.
- Ljung, G.M., Box, G.E.P. (1978). On a measure of lack of fit in time series models. *Biometrika*, 65(2), 297–303.
- Lo, A.W. (1991). Long-term memory in stock market prices. *Econometrica*, 59(5), 1279–1313.

References (5/7)

- Lo, A.W. (2004). The adaptive markets hypothesis. *Journal of Portfolio Management*, 30(5), 15–29.
- Lo, A.W. (2017). Adaptive Markets: Financial Evolution at the Speed of Thought. Princeton University Press.
- Lo, A.W., MacKinlay, A.C. (1988). Stock market prices do not follow random walks: evidence from a simple specification test. *Review of Financial Studies*, 1(1), 41–66.
- Lobato, I., Nankervis, J.C., Savin, N.E. (2001). Testing for autocorrelation using a modified Box–Pierce Q test. *International Economic Review*, 42(1), 187–205.
- MacKinlay, A.C. (1997). Event studies in economics and finance. *Journal of Economic Literature*, 35(1), 13–39.
- Malkiel, B.G. (2003). The efficient market hypothesis and its critics. *Journal of Economic Perspectives*, 17(1), 59–82.
- Martin, I.W.R., Nagel, S. (2022). Market efficiency in the age of big data. *Journal of Financial Economics*, 145(1), 154–177.
- McLean, R.D., Pontiff, J. (2016). Does academic research destroy stock return predictability?. *Journal of Finance*, 71(1), 5–32.
- Moskowitz, T.J., Ooi, Y.H., Pedersen, L.H. (2012). Time series momentum. *Journal of Financial Economics*, 104(2), 228–250.
- Nadarajah, S., Chu, J. (2017). On the inefficiency of Bitcoin. *Economics Letters*, 150, 6–9.
- Newey, W.K., West, K.D. (1987). A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica*, 55(3), 703–708.
- Pele, D.T., Lin, M.-B., Wang, R., Gaman, S. (2025). Market responses to ETH development milestones. IDA Institute Digital Assets, Bucharest University of Economic Studies.

References (6/7)

- Pele, D.T., Mazurencu-Marinescu, M. (2012). Modelling stock market crashes: the case of Bucharest Stock Exchange. *Procedia – Social and Behavioral Sciences*, 58, 533–542.
- Pele, D.T., Mazurencu-Marinescu, M., Nijkamp, P. (2013). Herding behaviour, bubbles and log periodic power laws in illiquid stock markets: a case study on the Bucharest Stock Exchange. SSRN Working Paper.
- Pele, D.T., Wesselhöfft, N., Härdle, W.K., Kolossiatis, M., Yatracos, Y.G. (2023). Are cryptos becoming alternative assets?. *European Journal of Finance*, 29(10), 1064–1105.
- Peng, C.-K., Buldyrev, S.V., Havlin, S., Simons, M., Stanley, H.E., Goldberger, A.L. (1994). Mosaic organization of DNA nucleotides. *Physical Review E*, 49(2), 1685–1689.
- Roberts, H.V. (1959). Stock-market "patterns" and financial analysis: methodological suggestions. *Journal of Finance*, 14(1), 1–10.
- Robinson, P.M. (1995). Gaussian semiparametric estimation of long range dependence. *Annals of Statistics*, 23(5), 1630–1661.
- Romano, J.P., Wolf, M. (2005). Stepwise multiple testing as formalized data snooping. *Econometrica*, 73(4), 1237–1282.
- Rozeff, M.S., Kinney, W.R. (1976). Capital market seasonality: the case of stock returns. *Journal of Financial Economics*, 3(4), 379–402.
- Samuelson, P.A. (1965). Proof that properly anticipated prices fluctuate randomly. *Industrial Management Review*, 6(2), 41–49; reprinted 2015, World Scientific.
- Shiller, R.J. (1981). Do stock prices move too much to be justified by subsequent changes in dividends?. *American Economic Review*, 71(3), 421–436 (NBER Working Paper 456).
- Shiller, R.J. (2026). U.S. stock markets 1871–present and CAPE ratio (monthly data, accessed September 2026).
- Shleifer, A., Vishny, R.W. (1997). The limits of arbitrage. *Journal of Finance*, 52(1), 35–55.

References (7/7)

- Stambaugh, R.F. (1999). Predictive regressions. *Journal of Financial Economics*, 54(3), 375–421.
- Sullivan, R., Timmermann, A., White, H. (1999). Data-snooping, technical trading rule performance, and the bootstrap. *Journal of Finance*, 54(5), 1647–1691.
- Summers, L.H. (1986). Does the stock market rationally reflect fundamental values?. *Journal of Finance*, 41(3), 591–601.
- Tak, R., Pele, D.T. (2025). LLM-driven stock prediction: capturing market trends with LLaMA. *Proceedings of the International Conference on Business Excellence*, 19(1), 529–540.
- Tak, R., Schlamp, S., Osterrieder, J., Pele, D.T. (2026). Racing the news: nanosecond market efficiency at Eurex and CME. SSRN Working Paper.
- Tran, V.L., Leirvik, T. (2020). Efficiency in the markets of crypto-currencies. *Finance Research Letters*, 35, 101382.
- Urquhart, A. (2016). The inefficiency of Bitcoin. *Economics Letters*, 148, 80–82.
- Urquhart, A., McGroarty, F. (2016). Are stock markets really efficient? Evidence of the adaptive market hypothesis. *International Review of Financial Analysis*, 47, 39–49.
- Wald, A., Wolfowitz, J. (1940). On a test whether two samples are from the same population. *Annals of Mathematical Statistics*, 11(2), 147–162.
- Welch, I., Goyal, A. (2008). A comprehensive look at the empirical performance of equity premium prediction. *Review of Financial Studies*, 21(4), 1455–1508.
- White, H. (2000). A reality check for data snooping. *Econometrica*, 68(5), 1097–1126.